

# Practice reshapes the geometry and dynamics of task-tailored representations

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Extensive practice makes task performance more efficient and precise, leading to automaticity. However, theories of automaticity differ on which levels of task representations (eg low-level features, stimulus–response mappings, or high-level conjunctive memories of individual events) change with practice, despite predicting the same pattern of improvement (eg power law of practice). To resolve this controversy, we built on recent theoretical advances in understanding computations through neural population dynamics. Specifically, we hypothesized that practice optimizes the neural representational geometry of task representations to minimally separate the highest-level task contingencies needed for successful performance. This involves efficiently reaching conjunctive neural states that integrate task-critical features nonlinearly while abstracting over noncritical dimensions. To test this hypothesis, human participants ( $n = 40$ ) engaged in extensive practice of a simple, context-dependent action selection task over 3 d while recording electroencephalogram (EEG). During initial rapid improvement in task performance, representations of the highest-level, context-specific conjunctions of task- features were enhanced as a function of the number of successful episodes. Crucially, only enhancement of these conjunctive representations, and not lower-order representations, predicted the power-law improvement in performance. Simultaneously, over sessions, these conjunctive neural states became more stable earlier in time and more aligned, abstracting over redundant task features, which correlated with offline performance gain in reducing switch costs. Thus, practice optimizes the dynamic representational geometry as task-tailored neural states that minimally tessellate the task space, taming their high dimensionality.

**Keywords:** abstraction; automatization; cognitive control; conjunctive representations; dynamics; EEG; geometry; skill learning.

## Introduction

Practice enhances performance on a wide range of tasks, from motor skills to more cognitively demanding, goal-directed behavior (Shiffrin and Schneider 1977; Newell 1993), like formulating geometry proofs (Neves and Anders 1980) or even writing books (Ohlsson 1992). Understanding how expertise in control-demanding tasks is achieved is crucial for explaining cognitive control processes, as it constrains how we perform goal-directed behavior following a controlled-to-automatic continuum.

Performance improvements over the course of practice are widely assumed to reflect changes in the neural representations of a task and the computations that access and act on those representations. Specifically, improvement is often thought to result from strengthening associations among task-relevant information, while reducing the influence of task-irrelevant factors, thereby facilitating access and use of task-relevant response pathways (Crossman 1959; Shiffrin and Schneider 1977; Mackay 1982; Cohen et al. 1990; Verguts and Notebaert 2009). For example, the dominance of word reading, as seen in the Stroop effect, is commonly attributed to strengthened associations between written color words and their spoken labels, developed through

extensive practice reading. In this way, reading becomes automatic when seeing a written word, allowing for highly efficient and accurate execution, even in the absence of attentional and control demands (Stroop 1935; Cohen et al. 1990; Moors and De Houwer 2006).

However, though many theories assume changes in task representations underlie practice, how and which representations change remain controversial. One reason for this fundamental gap in our understanding is that all tasks involve latent representations at multiple intermediate stages of computation, any or all of which could be the target of optimization (Wood and R nger 2016; Du et al. 2022; Zheng et al. 2024). Furthermore, the true objective functions of tasks are hard to infer even when the task structure is assumed to be fixed (Saxe et al. 2022). Accordingly, influential psychological theories of automaticity and skill learning, which comparably explain how cognitive control systems learn to reduce control and attentional demands while making performance more efficient and accurate, nevertheless differ substantively in terms of the level of task representations they assume change to improve performance.

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For example, explanations for the power function-like improvement (ie rapid improvement in the early stage of learning with diminishing benefits over practice) in task performance have been one of the fundamental targets of theories of automaticity and skill learning (Logan 1988; Newell 1990). Indeed, such a pattern of learning is so universally observed among tasks that it is among the few contemporary laws cited in psychology (Logan 1988; Teigen 2002). Yet, theories accounting for this phenomenon differ in the levels of representation they assume change with practice.

Newell and Rosenbloom's (1981) influential chunking theory proposed that practice yields chunking of task features at both intermediate and high levels of complexity and specificity, which collectively leads to heightened efficiency. Thus, power law-like gains in performance follow from the early benefits of learning the simpler intermediate chunks that generalize to multiple task events versus the complex chunks that are less frequently relevant. Other theories similarly localize learning at an intermediate compositional level and, most commonly, in terms of context-independent stimulus-response (S-R) bindings (Schneider 1985; Cohen et al. 1990).

In contrast, Logan's (1988) instance theory of automaticity assumes that improvement reflects a shift of behavioral control from a slow but flexible algorithmic process to a one-step direct retrieval of specific, high-level task-event instances that combine all task features, including the context. Practice adds instances to memory, which facilitates this retrieval process. In time, retrieval predominates but with diminishing returns on faster performance. Thus, unlike the above cases, learning occurs only at the highest level (ie memory of individual events) and not at intermediate levels.

Thus, even for the most widely observed learning effects, multiple theories explain the same pattern of learning outcomes using different psychological constructs. Hence, a theoretically important question concerns the locus of representational change during practice that leads to behavioral improvement. What task representations are targeted by practice?

One approach to addressing this question is characterizing how practice optimizes the representational geometry of neural task representations and how these changes translate to improved performance (Kriegeskorte and Kievit 2013; Musslick et al. 2020; Saxe et al. 2022; Johnston and Fusi 2023; Pang et al. 2023; Li et al. 2024). Theoretically, a geometric view of task representations could unify selective enhancement, reduction (or abstraction), optimization, and the underlying mechanisms of information coding and neural changes within a coherent framework (Kriegeskorte and Kievit 2013; Fusi et al. 2016; Musslick et al. 2020; Badre et al. 2021; Duncker and Sahani 2021; Farrell et al. 2022; Saxe et al. 2022). This perspective has the potential to bridge constructs in psychological theories (eg chunks and instances) to their neural computations. Yet, empirical evidence on characterizing how practice reshapes neural task representations over learning trajectories is still scarce (Mill and Cole 2023; Wojcik et al. 2023).

Accumulating evidence suggests that the specificity and generalizability of task representations adapt to the task demands by shaping their geometry and dimensionality (Fusi et al. 2016; Badre et al. 2021; Jazayeri and Ostojic 2021; Li et al. 2024), which are both important outcomes of practice. In general, high-dimensional task representations are associated with increased separability of task features and more efficient and accurate task performance. This is typically achieved by mixing separate coding axes for the combinations of task features in a structured manner (Bhandari et al. 2024; Fine et al. 2022; Ito et al. 2022; Kikumoto et al. 2024; Kikumoto and Mayr 2020; Ritz and Shenhav 2024; Tang et al.

2020; Weber et al. 2023), such as in a nonlinear conjunction of all task-relevant features (stimuli, responses, and contexts), that define a task event, or in a random and task-agnostic manner (Rigotti et al. 2013, 2010; Tang et al. 2019). In contrast, low-dimensional task representations typically benefit generalization or re-use of neural solutions. This is because task representations are abstracted over task-irrelevant features to generate compositional manifolds (Hazy et al. 2007; Bernardi et al. 2020; Flesch et al. 2022; Vaidya and Badre 2022; Courellis et al. 2023; Goudar et al. 2023; Johnston and Fusi 2023; Driscoll et al. 2024; Tafazoli et al. 2024) or compressed representations (Hummos et al. 2022; Muhle-Karbe et al. 2023).

What is unknown is how task representations evolve to these task-tailored neural states over the course of learning, particularly in tasks where nonlinear integrations of contexts with response pathways are required for successful performance (but see Courellis et al. 2023; Mill and Cole 2023; Wojcik et al. 2023). On one view, new tasks are initially computed compositionally, and, over time, specific combinations are learned as context-specific, high-dimensional representations. To some extent, this change mirrors predictions by some theories of automaticity described above, in that progress to highly skilled behavior increasingly relies on a library of specific representations (eg chunks or instances). Conversely, however, early in task learning, expressive high-dimensional representations might make multiple readouts available for optimization (Farrell et al. 2023; Flesch et al. 2022; Tang et al. 2019). Over the course of practice, irrelevant subspaces could be compressed or aligned to avoid interference or redundancy (Cohen et al. 1990). This results in a transition toward a compositional and low-dimensional geometry. As a third alternative, behavioral improvement may occur strictly at the handling of sensory and/or motor information or maintenance of contexts (Anderson 1982; Cole et al. 2018; Meyer and Kieras 1997; Schumacher et al. 2001).

Thus, there is a significant empirical gap in our understanding of how the representational geometry of task representations changes in human brains throughout practice in ways that yield behavioral improvements. This issue remains unresolved because various theoretical constructs can equally well account for the same learning effects on behavior without clarifying underlying neural task representations. The current study addresses this gap in knowledge by directly relating changes in the geometry and dynamics of task representations to improvement in performance over 3 d of context-dependent action selection. Participants performed tasks requiring the nonlinear integration of rule context to select appropriate S-R mappings, shared across contexts (Mayr and Bryck 2005). This setup allowed us to explore the learning of many task representations of varying complexity, including S-R mappings that are context-independent or context-specific (ie task-event file; Hommel 2004; Verguts and Notebaert 2009; Schumacher and Hazeltine 2016; Frings et al. 2020). Further, we mapped multiple task cues to the same rule context, which allowed us to test how such redundant task features become accommodated over practice.

We hypothesized that expertise in goal-directed actions arises from the optimization of the representational geometry in a task-tailored manner. Specifically, neural states maximally separating required context-specific task contingencies should be enhanced because they provide an efficient readout to solve the targeted task. Such neural states are equivalent to encoding context-specific conjunctions—nonlinear integration of task features such as the context and S-R mapping to define a task event. In other words, practice should enhance stable readout of context-specific conjunctive neural states to improve task performance.

Further, we hypothesized that these separable, conjunctive subspaces are gradually tailored to align unnecessary dimensions to abstract task features that are not critical to solving the task at hand because such solutions are minimally high in dimensionality.

Consistent with these hypotheses, we found that practice optimizes the representational geometry of task representations selectively. Specifically, context-specific, high-level conjunctive representations were present early in practice and became stronger as a function of power law, selectively accounting for rapid performance improvements, which was not the case for other lower levels of representation. Across days, these conjunctions recruited more separable yet stable dynamics at earlier points relative to the start of the trial. At the same time, these dynamics were reshaped to become more abstract across redundant task features, reducing the unnecessary dimensions of tasks. Overall, these findings suggest that practice optimizes encoding and/or retrieval of task-tailored, context-specific conjunctions. These conjunctions minimally orthogonalize contingencies to solve the task and slowly become abstract over redundant task features, serving as the locus of practice leading to expertise.

## Materials and methods

### Participants

Forty participants (20 females; mean age: 23 years) were recruited and gave informed consent using procedures approved by the Human Subjects Committee at the RIKEN. All participants had normal or corrected-to-normal vision and had no history of neurological or psychiatric disorders. After preprocessing the EEG data, one participant was removed and was not analyzed further due to excessive artifacts (ie more than 25% of trials; see [EEG Recordings and Preprocessing](#) for details). Two participants did not complete the last session, but their data for earlier sessions were retained for the analysis.

### Behavioral procedure

The task closely followed the procedure in [Kikumoto and Mayr \(2020\)](#). Specifically, participants performed a cued rule-based action selection task ([Fig. 1A](#)), which required participants to behave according to 16 unique rule-S-R conjunctions ([Fig. 1B](#)). On a trial-by-trial basis, a randomly selected action rule (ie vertical, horizontal, clockwise, and counter-clockwise rule) specified four different S-R mappings, based on a simple spatial transformation over four possible stimuli (a dot at the top left, top right, bottom left, and bottom right of the white frame). The spatial transformation indicated the target button-press response from four options that were arranged in a  $2 \times 2$  square matrix (4, 5, 1, and 2 on the number pad) that mirrored the stimulus display. Thus, for instance, the “vertical” rule mapped the top-left circle to the bottom-left response. Two different synonymous cue words were randomly used to specify action rules. Further, though the sets of four rule-S-R conjunctions were unique for a given rule, individual S-R mappings could be shared across different rules. For example, just as with the vertical rule, the counter-clockwise rule would also map a circle in the top-left to the bottom-left response, but this rule would require a different S-R mapping than the vertical rule for a circle in the upper left or bottom right. Following the logic of [Mayr and Bryck \(2005\)](#), this design feature allowed us to distinguish the effects of practice on context-integrated conjunctive representations versus shared, compositional stimulus-response rules.

Participants were tested in three sessions across three consecutive days ([Fig. 2](#)). Each session consisted of 202 experimental

blocks, where the first two blocks did not provide performance-based incentives but were identical to other blocks. On the first day, participants practiced 5 blocks without EEG measurement. Participants were instructed to complete as many correct trials as possible that commenced within each 18-s block (trials that began within the 18-s window were allowed to finish). Participants were given a performance-based incentive for trials with response times (RTs) faster than the 75th percentile of correct responses in the preceding blocks when 1) the overall accuracy was above 90% and 2) there were more than 7 completed trials in a given block.

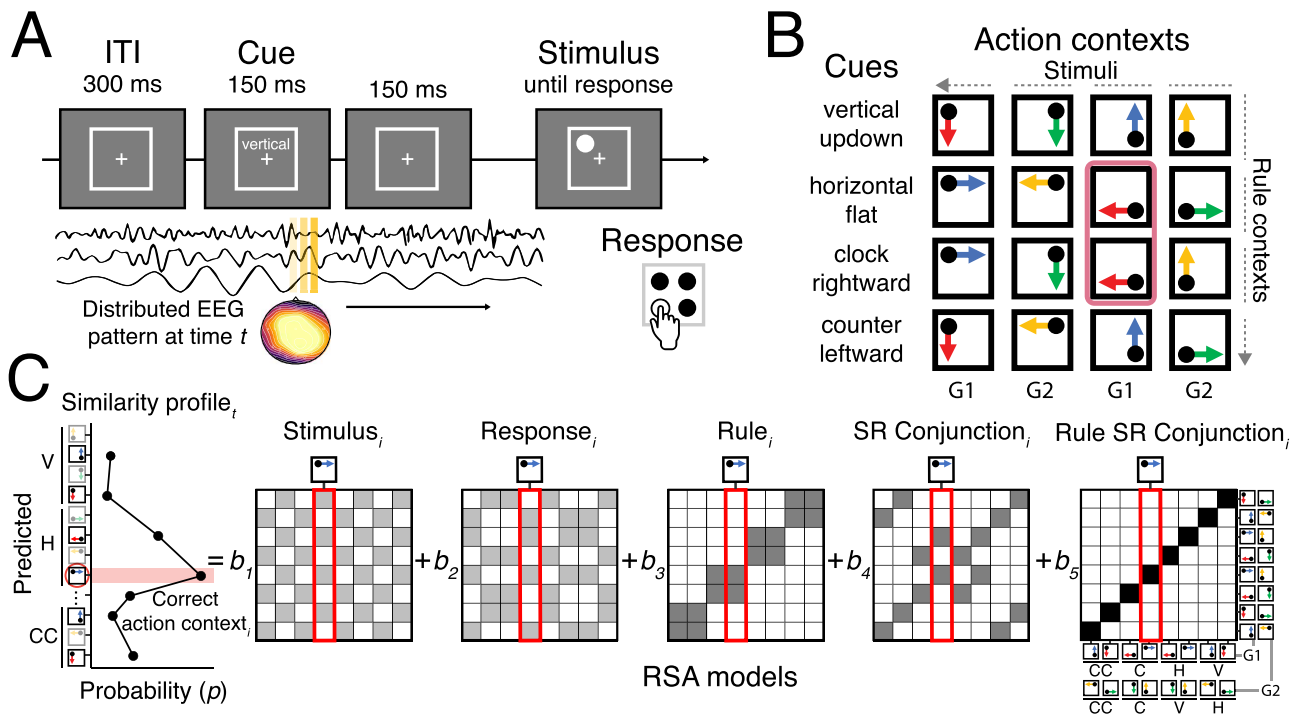
### EEG recordings and preprocessing

The continuous EEG was recorded using a Brain Products actiCHamp recording system (Brain Products GmbH). Recordings were obtained from a broad set of scalp sites (Fp1, Fp2, F7, F3, F4, F8, FC5, FC1, FC2, FC6, T7, C3, C4, T8, CP5, CP1, CP2, CP6, P7, P3, P4, P8, PO9, PO10, O1, O2, Fz, FCz, Cz, Pz, and Oz), from the left and right mastoids, and from electrodes lateral to the external canthi and below the left eye. The electrodes (TP9 and TP10) were placed on the mastoids beneath the cap as reference sites. Using additional passive electrodes, the electrooculogram (EOG) was recorded from laterally placed electrodes to detect horizontal eye movements with the separate ground electrode placed on the left side of the forehead. The EEG was filtered online with 0.1 Hz high-pass and 1000 Hz low-pass filters and digitized at 1000 Hz. The left mastoid was used as a reference for all recording sites, and data were re-referenced off-line to the average of left and right mastoids.

The scalp EEG and EOG were amplified with an SA Instrumentation amplifier with a bandpass of 0.01 to 0.45 Hz, and signals were downsampled at 250 Hz using EEGLab ([Delorme and Makeig 2004](#)). EEG data were first segmented into 22-s intervals to include all trials within a block. After time-frequency decomposition was performed, these epochs were further segmented into smaller epochs for each trial. For stimulus-aligned epochs, we used the time interval of  $-200$  to  $800$  ms relative to the onset of stimuli. For response-aligned epochs, we used the time interval of  $-800$  to  $200$  ms relative to the onset of responses. The trial-to-trial epochs including blinks ( $>250 \mu\text{V}$ , window size =  $200$  ms, window step =  $50$  ms), large eye movements ( $>25 \mu\text{V}$ , window size =  $200$  ms, window step =  $10$  ms), blocking of signals (range =  $-0.01$  to  $0.01 \mu\text{V}$ , window size =  $200$  ms) were excluded from subsequent analyses. Any epochs that showed EEG activity greater than 5 s.d. were excluded.

### Time-frequency analysis

Temporal-spectral profiles of single-trial EEG data were computed via complex wavelet analysis ([Cohen 2014](#)) by applying time-frequency analysis to preprocessed EEG data epoched for the entire block ( $>22$  s to exclude the edge artifacts). The power spectrum was convolved with a series of complex Morlet wavelets ( $e^{i2\pi ft} e^{-(t/\sigma)^2/2}$ ), where  $t$  is time,  $f$  is frequency increased from 2 to 40 Hz in 35 logarithmically spaced steps, and  $\sigma$  defines the width of each frequency band, set according to  $n/2\text{pft}$ , where  $n$  increased from 3 to 10. We used logarithmic scaling to keep the width across frequency bands approximately equal, and the incremental number of wavelet cycles was used to balance temporal and frequency precision as a function of frequency of the wavelet. After the convolution was performed in the frequency domain, we took the inverse of the Fourier transform, resulting in complex signals in the time domain. A frequency band-specific estimate at each time point was defined as the squared magnitude of the convolved signal for instantaneous power.



**Fig. 1.** Task design and the procedure of decoding analysis. Both the task design and the decoding analysis procedure are identical to Exp.2 in Kikumoto and Mayr (2020). A) Sequence of trial events in the rule-selection task. B) Spatial translation of different rules (rows) mapping different stimuli (columns) to responses (arrows), yielding 16 independent conjunctions. Two action contexts highlighted, for example, share the S-R mapping. C) Schematic of the time-resolved representational similarity analysis (RSA). For each sample time ( $t$ ), a scalp-distributed pattern of EEG was used to decode the specific rule/stimulus/response configuration of a required action. The decoder produced sets of classification probabilities for each of the possible action contexts. The profile of classification probabilities reflects the similarity structure of the underlying representations, where action contexts with shared features are more likely to be confused. For each trial and timepoint, the classification probabilities were regressed onto model vectors as predictors that reflect the different possible representations. In each model matrix, the shading of squares indicates the theoretically predicted classification probabilities (darker shading means higher probabilities) in all possible pairs of action contexts. The coefficients associated with each predictor reflect the unique variance explained by each of the constituent features and two different types of conjunctions: stimulus–response (S-R) conjunction and rule-specific S-R (Rule S-R) conjunction. To fully counterbalance the conditions, action contexts were grouped into two (G1 and G2), and then, the RSA procedure was applied.

## Representational similarity analysis

We performed a time-resolved multivariate pattern classification analysis to decode action-relevant information following our previously reported method with a few modifications (Kikumoto and Mayr 2020). As the first step, at every sample in trial-to-trial epochs, separate linear decoders were trained to classify all possible action contexts (Fig. 1C) for each S-R mapping. Specifically, we performed a penalized linear discriminant analysis using “mda: Mixture and Flexible Discriminant Analysis” package developed by Hastie and colleagues (Hastie et al. 2024), integrated with the “caret” package in R (Hastie et al. 1995; Kuhn 2013). The hyperparameter of the penalized linear discriminant analysis is a lambda parameter, which is the shrinkage penalty coefficient applied to the covariance matrix of trained features for the ridge regression during estimation of optimal scaling. A lambda parameter yielding the highest decoding accuracy was estimated in the training sets and then tested on the independent test sets for cross-validation.

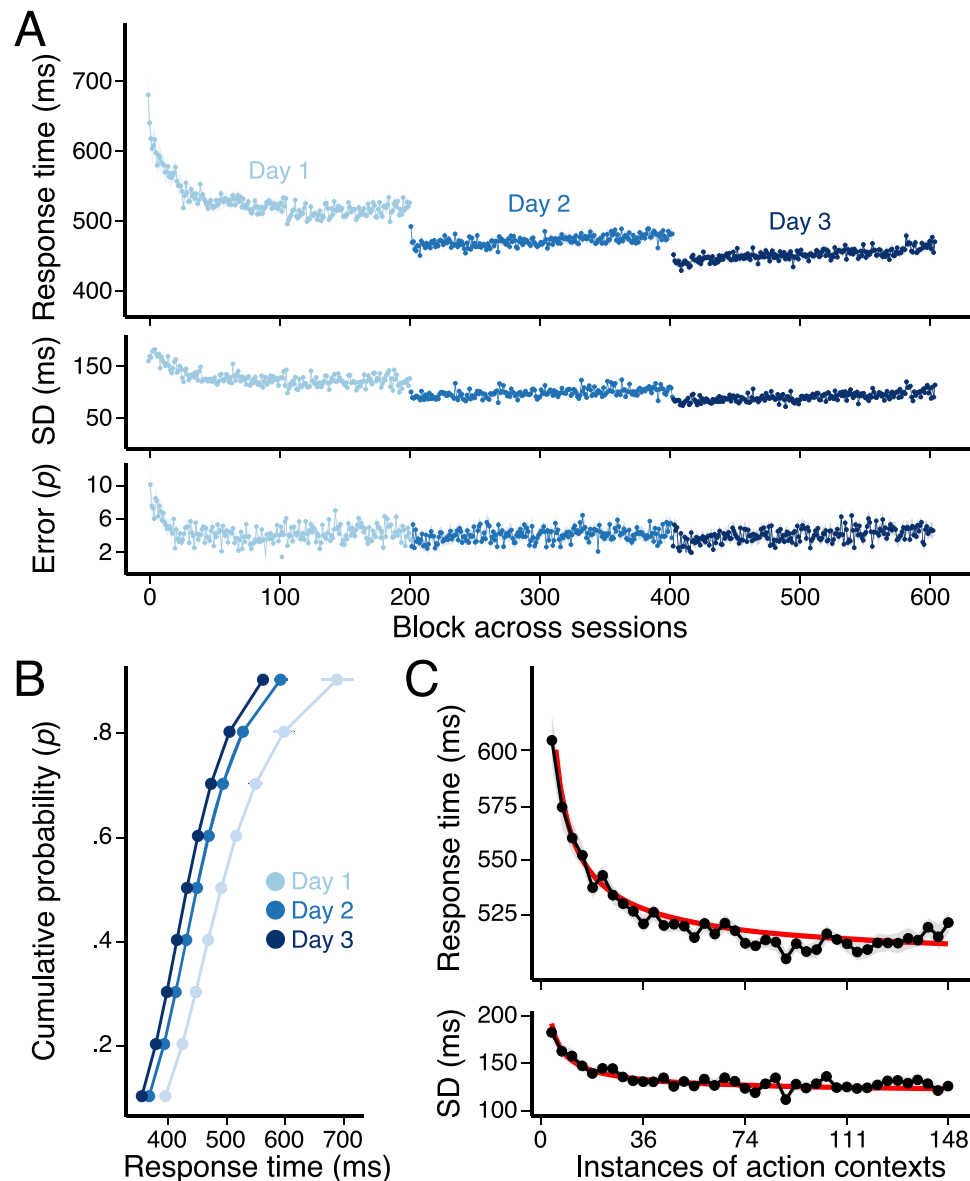
Decoders were trained with the instantaneous power of rhythmic EEG activity, which was averaged within the predefined ranges of frequency values (2 to 3 Hz for the delta band, 4 to 7 Hz for the theta band, 8 to 12 Hz for the alpha band, 13 to 30 Hz for the beta band, 31 to 35 Hz for the gamma band), corresponding to 155 features (5 frequency bands  $\times$  31 electrodes) to learn multiclass classifications. This step produced a graded profile of classification probabilities for each action context, reflecting

the similarity of underlying multivariate neural patterns between actions.

Prior to the training of decoders, trials with incorrect responses were excluded. Within individuals and frequency bands, the data points were z-transformed across electrodes to remove effects that scaled all electrodes uniformly. We used a  $k$ -fold repeated, cross-validation procedure to evaluate the decoding results (Mosteller and Tukey 1968) by randomly partitioning single-trial EEG data into five independent folds with an equal number of observations of each action context. After all folds served as the test sets, each cross-validation cycle was repeated 10 times, in which each step generated a new set of randomized folds. The number of observations in each target class (eg action contexts or a set of task conditions defined by rule, stimulus, and response features) was equated by randomly dropping excess trials for certain conditions. Resulting classification probabilities were averaged across all cross-validated results with the best-tuned hyperparameter.

We next performed representational similarity analysis (RSA) on the graded classification probabilities to assess the underlying similarity structure of task variables. The specific set of four rules ensured that each S-R mapping occurred in two different rules (eg a top-left circle leads to a bottom-left response in both the vertical and the clockwise rule), allowing us to decode context-specific S-R conjunctions and rule-independent S-R conjunctions separately (Fig. 1B; Kikumoto et al. 2022a). Thus,





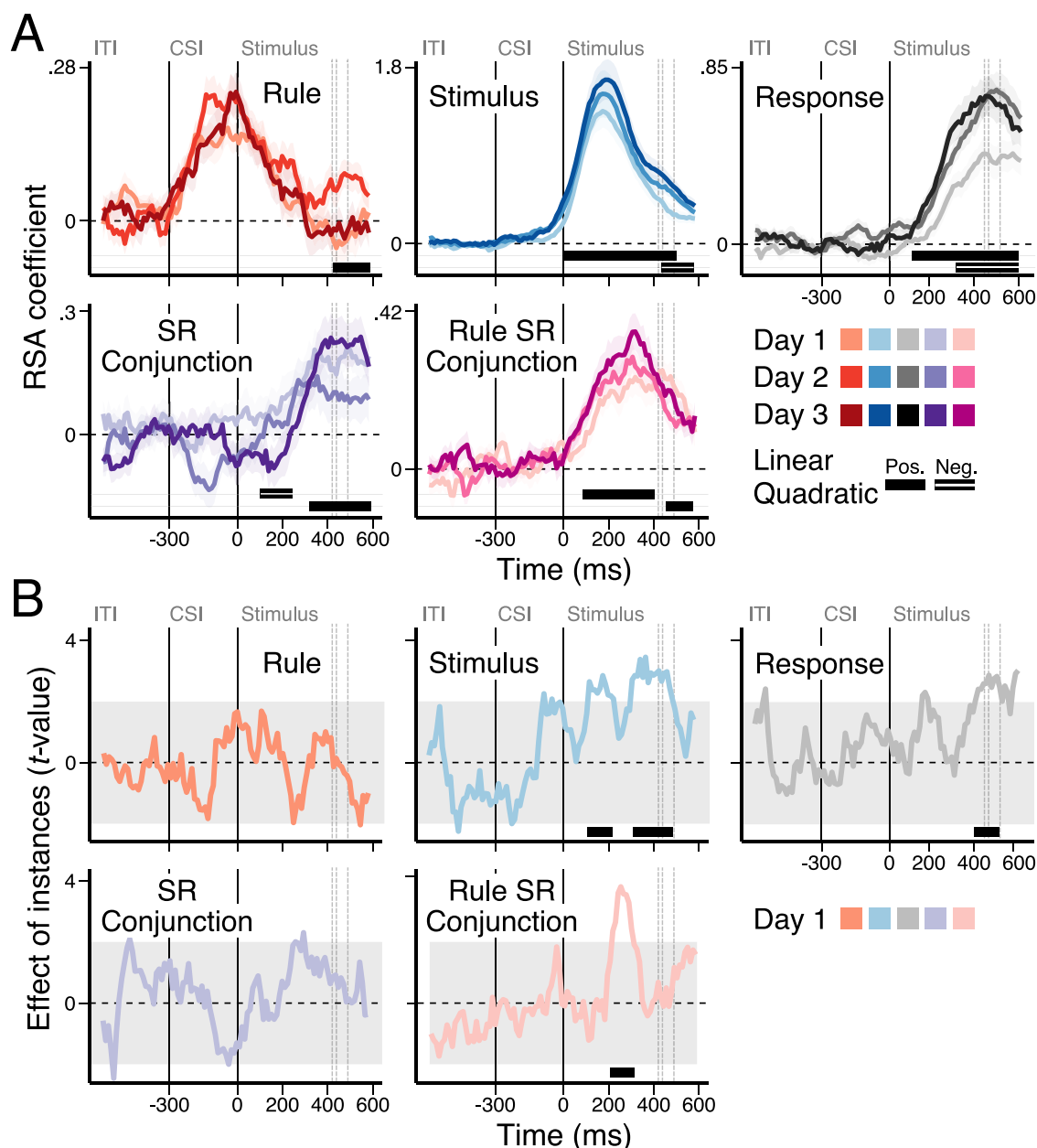
**Fig. 2.** Practice effects on task performance. A) The average response times (RTs), the variability (standard deviations or SDs) of RTs, and the probability of errors within each block. B) Changes in RTs across days over the cumulative density of RT distributions on each day. C) Changes in RTs and SDs on day 1 sorted by the binned instances of specific action contexts (Fig. 1B). Each bin contains on average 40 trials. Lines show the best-fitted power functions with the same exponent.

each RSA model matrix represents the distinct similarity structure arising from different hypothetical, underlying representations (eg rules, stimuli, responses, S-R conjunctions, and context-specific S-R conjunctions; Fig. 1C).

To obtain time-resolved estimates of each RSA model, we regressed the vector of log-transformed classification probabilities onto RSA model vectors for independent decoding results. Importantly, to estimate the unique variance explained by competing models, we regressed all model vectors simultaneously. In the current design, only the consistent task features such as rule, stimulus, and responses are orthogonally factorized, and their conjunctions are, by definition, correlated to other task features (Supplementary Fig. 1A). To fully counterbalance conditions, models were fitted within subgroups of action contexts (ie G1 and G2; Fig. 1B), following the procedure in Kikumoto and Mayr (2020). Using simulated data, we verified that single-trial

RSA with subgroups of action contexts, including both context-independent (S-R) conjunctions and context-dependent (rule-S-R) conjunctions, can be estimated (Supplementary Fig. 1B).

Further, individual time samples are concatenated within 12-ms nonoverlapping time windows. The nuisance regressors were included at the stage of single-trial RSA where RSA model vectors were regressed onto the graded classifier probabilities. Each subject had two nuisance regressors: the average RT and errors in 8 unique action contexts for G1 and G2 subgroups of action contexts. These procedures allow us to prevent the inflation of decodability driven by superfluous factors unrelated to trial-to-trial conditions or behavioral performance. These coefficients, expressed in their corresponding t-values, reflect the quality of action representations at the level of single trials, which was later related to variability in behavior and dimensionality measures. We excluded t-values that exceeded 5 SDs from the mean of



**Fig. 3.** Stimulus-aligned time course of decoding of task representations. A) Average, single-trial RSA coefficients (t-values) associated with each of the basis set task features (rule, stimulus, and response) and two conjunctions (S-R conjunction and rule-specific S-R conjunction) that are aligned to the onset of the stimulus. Lines within each panel show the average RSA scores on each day. Shaded regions specify within-subject standard errors. B) Modulation of RSA scores by the accumulated experiences of action context instances. A, B) The black bars at the bottom of each panel denote the significant clusters using a nonparametric permutation test (cluster-forming threshold,  $P < 0.05$ , cluster-significance threshold,  $P < 0.01$ , two-tailed). Gray dotted lines show the mean RTs for each day.

each sample point. This procedure excluded less than 1% of all samples. These analysis steps were repeated separately using stimulus-aligned (Figs. 3 and 4) and response-aligned EEG signals (Supplementary Fig. 2).

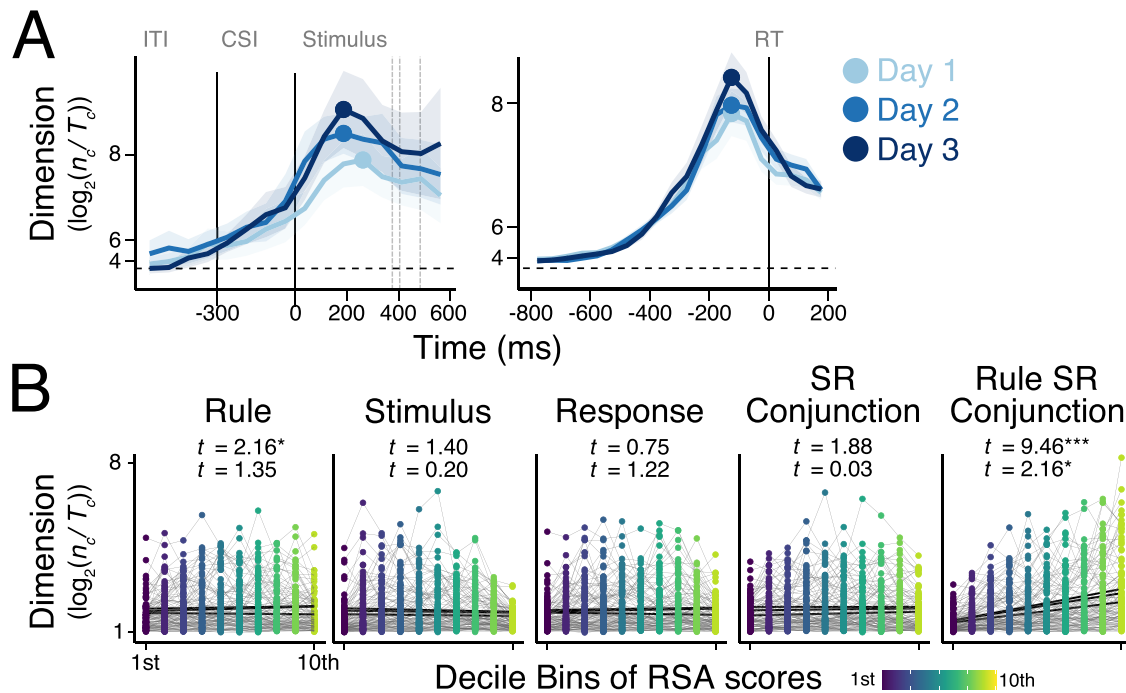
### Multilevel modeling

We used multilevel models to further examine the decoded representations (Fig. 3B; Table 1) and their relationship to trial-by-trial variability in behavior.

When analyzing effects specific to particular action contexts (eg learning effects on day 1), we specified two levels of random effects in the models: individual action contexts (16 unique task conditions in Fig. 1B) and participants. In analyses where the

focus is session-level effects, we dropped the level of action contexts. RTs were log-transformed prior to analysis. To assess RT variability, we used the standard deviation of RTs within individual blocks (Fig. 2A) or within bins of accumulated instances of individual action contexts (Fig. 2C) as dependent variables. For all multilevel models, we included both the random intercepts and the random slopes at the level of participants by default, unless stated otherwise.

For the time-resolved regression (Fig. 3B), we used nonparametric permutation tests to evaluate the decoding results in the time domain (Maris and Oostenveld 2007). First, we performed a series of regression analyses over time and detected samples that exceeded the threshold for cluster identification



**Fig. 4.** Task representational dimensionality. A) Average dimension score during action selection. The dimensionality is estimated as the proportion of implementable binary separation,  $n_c/T_c$ , out of all possible arbitrary binary pairs. The left and right panels correspond to the results using signals aligned to the onset of stimulus or responses, respectively. Shaded regions specify within-subject standard errors. B) Changes in the representational dimensionality as a function of RSA scores of each of the basis set task features (rule, stimulus, and response) and their conjunctions (S-R conjunctions and rule S-R conjunction). The dimension scores ( $n_c/T_c$ ) are calculated separately in decile bins of RSA scores of each feature within subjects, which correspond to individual points in each panel. The two t-values within each panel are the statistic of a linear effect of an increase in RSA scores (top) and how it changes across days (bottom).

**Table 1.** MLM testing the effects of sessions (days) on task representations during early response selection (0 to 300 ms from the stimulus onset).

| Effects             | Rule            |         | Stimulus         |         | Response        |         | S-R Conjunction  |         | Rule-specific Conjunction |         |
|---------------------|-----------------|---------|------------------|---------|-----------------|---------|------------------|---------|---------------------------|---------|
|                     | beta (CI)       | t-value | beta (CI)        | t-value | beta (CI)       | t-value | beta (CI)        | t-value | beta (CI)                 | t-value |
| Session (linear)    | <.01 (-.01 .01) | .54     | .03 (.01 .06)    | 2.70    | .02 (.01 .04)   | 3.29    | -.02 (-.03 -.01) | -2.12   | .02 (.01 .04)             | 2.34    |
| Session (quadratic) | <.01 (-.01 .01) | -.71    | <-.01 (-.02 .02) | -.08    | -.01 (-.02 .01) | -1.37   | -.01 (-.01 .01)  | -.43    | -.01 (-.01 .01)           | -.81    |

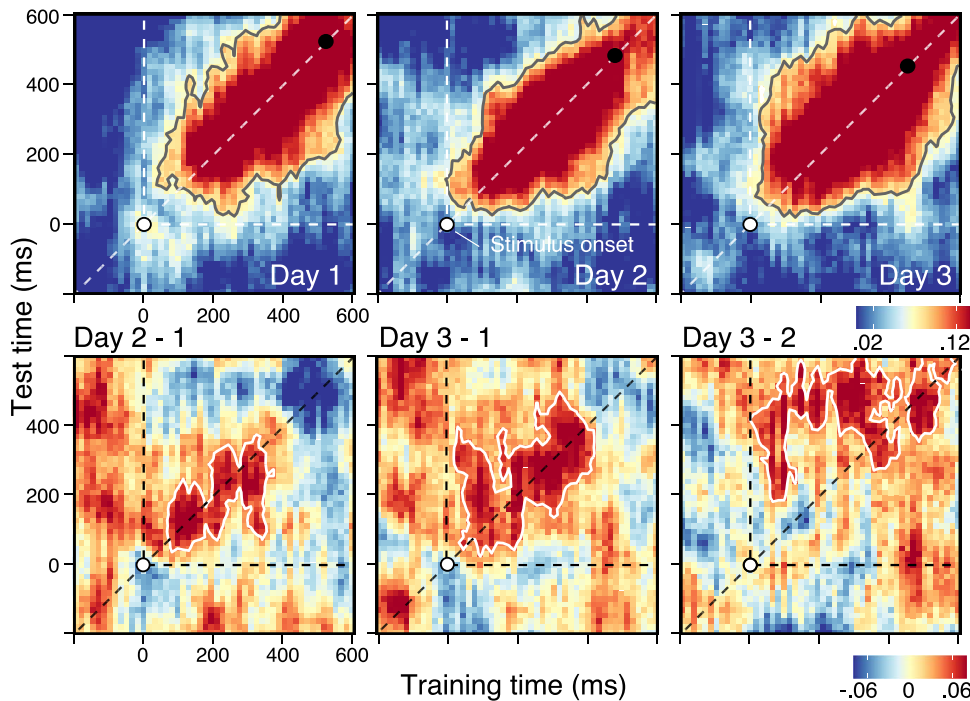
(cluster-forming threshold,  $P < 0.05$ ). Then, empirical cluster-level statistics were obtained by taking the sum of t-values in each identified cluster with consecutive time points. Finally, nonparametric statistical tests were performed by computing a cluster-level P-value (cluster-significance threshold,  $P < 0.05$ , two-tailed) from the distributions of cluster-level statistics, which were obtained by Monte Carlo iterations of regression analysis with shuffled condition labels. For models with RSA scores averaged over time windows, the RSA scores of action features were averaged over time intervals selected a priori based on previous studies: precue period (−300 to 0 ms from the onset of stimulus), early response selection period (0 to 300 ms relative to the onset of stimulus), and late response selection period (−300 to 0 ms relative to the execution of responses). These time windows were selected based on our previous studies (Kikumoto and Mayr 2020; Kikumoto et al. 2022a; Kikumoto et al. 2022b).

### Temporal generalization analysis

To assess the temporal stability of task representations, we used the temporal generalization analysis (Dehaene and King 2016). First, we trained a set of decoders to classify action contexts, following the identical procedure as the time-resolved RSA analysis, using EEG patterns at the specific time point. These

time-specific decoders were applied to the held-out data obtained at identical (matched) or different (generalized) time intervals. This generates a series of decoding results in a two-dimensional matrix form where the on-diagonal entries correspond to the results in matched time points whereas the off-diagonal entries are the results of generalization of linear hyperplanes over other time. Assessing the pattern of temporal generalization over the course of learning reveals how the nature of underlying neural computation such as the temporal stability of linear readouts of task-critical information changes.

The resulting decoding results, corresponding to every pairing of training and testing nonoverlapping time intervals (window size = 8 ms), were submitted to the regression-based RSA procedure to decode rules, stimulus, response, S-R conjunctions, and context-specific conjunctions simultaneously. Both the dynamic and stable components were defined in the prespecified time window (ie 0 to 300 ms from the stimulus onset) by averaging RSA coefficients from the subset of results in the cross-time generalization matrix (Fig. 5). The dynamic component was defined by the on-diagonal results, where the timings of training and test data matched. In contrast, the stable component was defined by the off-diagonal results, where the timings of the training and test data differed. To capture the early stabilization of representations, the generalization in forward time was used.



**Fig. 5.** Representational dynamics in the subspace of conjunctive control representations. Temporal generalization matrix of linear hyperplanes that separate rule-specific conjunctions. The x axis denotes time points where signals were sampled to train decoders and then submitted to single-trial RSA. The y axis denotes time points where the test sets were derived. The regions surrounded by a gray/white contour denote the significant clusters using a nonparametric permutation test (cluster-forming threshold,  $P < 0.05$ , cluster-significance threshold,  $P < 0.01$ , two-tailed). The top panels show changes across days (black dots mark the average RTs), and the bottom panels show the differences between consecutive days. For the difference in temporal stabilities across days (bottom row panels), the cluster-based permutation was performed within time windows where significant session-wise effects were detected.

## Dimensionality analysis via binary classification method

To assess the representational dimensionality (ie shattering dimension) of task representations, we used a binary (pairwise) classification method (Rigotti et al. 2013; Bernardi et al. 2020), which evaluates the dimensionality by counting how many different input conditions (ie unique action contexts; Fig. 1B) could be reliably separated or shattered into binary groupings by linear readouts ( $n_c/T_c$ ). We adapted the original binary classification method introduced in Rigotti et al. (2013) to EEG data that contain a non-trivial level of noise by adding constraints on how we select implementable binary classifications (Kikumoto et al. 2024).

Specifically, we trained separate decoders for all possible 65,536 (ie  $2^{16}-2$ , excluding two cases where all labels being identical) sets of arbitrary binary separations defined by 16 unique input conditions (Fig. 1B). Resulting decoding accuracies were counted as reliable linear separations when the following two conditions were satisfied: (i) the decoding accuracies exceeded the arbitrary chosen cutoff threshold of decoding accuracy of 0.51 where the chance level is 0.50, and (ii) above-threshold decoding accuracies were observed in each of the unique input conditions exclusively. This second criterion was applied to increase the sensitivity to detect changes in dimensionality scores driven by the nonlinear mixing of information even when the cutoff threshold accuracy is significantly lowered to the chance level. Note that the effect of different cutoff method was tested previously (Kikumoto et al. 2024), which did not affect differences in dimensionality across compared conditions in empirical data.

As a result, we obtained the proportion of reliable binary separations out of all possible binary groupings ( $n_c/T_c = \text{count}$

of implementable binary classifications, which approximates the dimensionality of neural responses to the input space [dimension =  $\log_2(n_c/T_c)$ ]. When comparing across days, the aforementioned exclusive cutoff was applied separately. The other settings of decoding analyses such as the steps of cross-validation and the exclusion criteria of trials were identical to the RSA analysis except that EEG signals were averaged over 50 ms nonoverlapping consecutive time windows to reduce the number of interactions.

## Results

### Practice effects on behavior

Within the first session of the task (Fig. 1), the average RT, error rates, and variability of RT (ie standard deviations of RT in each experimental block) decreased over experimental blocks and approached an asymptote, following a power-like function. These changes were reliably explained by both linear and quadratic effects (Fig. 2A; mean RT:  $t(1,38) = -5.57$ ,  $\beta = -0.41$ , 95% CI  $[-0.56 -0.28]$  for linear effect, and  $t(1,38) = 7.02$ ,  $\beta = 0.32$ , 95% CI  $[0.23 0.40]$  for quadratic effect; error rate:  $t(1,38) = -1.16$ ,  $\beta = -0.31$ , 95% CI  $[-0.79 0.20]$  for linear effect, and  $t(1,38) = 4.80$ ,  $\beta = .91$ , 95% CI  $[0.54 1.28]$  for quadratic effect; variability of RT:  $t(1,38) = -3.64$ ,  $\beta = -0.93.16$ , 95% CI  $[-1.43 -0.43]$  for linear effect, and  $t(1,38) = 5.63$ ,  $\beta = 0.84$ , 95% CI  $[0.54 1.13]$  for quadratic effect).

To investigate learning trajectories at the early stage of learning within each action context, we sorted trials into 40 bins by counting cumulative encounters (ie instances) of correctly executed responses in action contexts corresponding to the occurrence of a specific rule-S-R conjunction. Both RT and the variability of RT decreased as the number of experienced



instances of specific conjunctions increased (Fig. 2C; Mean RT  $t(1,38) = -4.21$ ,  $\beta = -0.21$ , 95% CI  $[-0.30 -0.11]$  for linear effect, and  $t(1,38) = 6.62$ ,  $\beta = 0.17$ , 95% CI  $[0.12 0.22]$  for quadratic effect; Variability of RT:  $t(1,38) = -1.92$ ,  $\beta = -0.26$ , 95% CI  $[-0.53 0.01]$  for linear effect, and  $t(1,38) = 4.62$ ,  $\beta = 0.31$ , 95% CI  $[0.18 0.44]$  for quadratic effect). These results indicate that task performance is rapidly optimized within the first session among independent action contexts in accord with the expected power-law pattern.

The offline gain (ie consolidation) in task performance tends to diminish for later sessions (Karni et al. 1994; Newell and Rosenbloom 1981). Based on these previous studies on skill learning, we expected that gains in task performance would deflate for later sessions. We include a quadratic term to capture this expected decelerating component. Comparing the first, second, and third days, both the average RT and variability of RT substantially decreased whereas error rates remained near floor (Fig. 2A; Mean RT:  $t(1,38) = -6.40$ ,  $\beta = -0.15$ , 95% CI  $[-0.20 -0.10]$  for linear effect, Error rate:  $t(1,38) = 7.28$ ,  $\beta = -0.24$ , 95% CI  $[-0.30 -0.17]$  for quadratic effect;  $t(1,38) = -1.84$ ,  $\beta = -0.02$ , 95% CI  $[-0.05 0.01]$  for linear effect, variability of RT: and  $t(1,38) = -0.72$ ,  $\beta = -0.01$ , 95% CI  $[-0.05 0.02]$  for quadratic effect for error rates;  $t(1,38) = -6.40$ ,  $\beta = -0.15$ , 95% CI  $[-0.20 -0.10]$  for linear effect and  $t(1,38) = -7.28$ ,  $\beta = -0.24$ , 95% CI  $[-0.30 -0.17]$  for quadratic effect). The linear improvements in RTs were observed in all steps of the vincentized RT distributions (Fig. 2B). Though, within the later sessions, performance became significantly worse as the experimental session progressed [ $t(1,38) > 4.43$ ], possibly due to fatigue. Nevertheless, these results indicate that after initial rapid improvement, task performance gradually improved over sessions. Because such learning effects are apparent at the beginning of later sessions, this improvement was due to task offline gains between sessions.

Next, we assessed the patterns of repetition priming costs in switching rule, stimulus, and response feature dimensions across trials. A lack of repetition benefits under switching of any task features (ie partial overlap of task features) across trials has been considered evidence of task feature binding (ie event-files: (Hommel 2004; Mayr and Bryck 2005; Frings et al. 2020; Rangel et al. 2023) and has been linked to the formation of conjunctions integrating goals or contexts (Kikumoto and Mayr 2020). Consistent with these studies, we found that switching in any task feature dimension incurred substantial costs on behavior compared to trials when the rule, stimulus and response repeats completely, (Supplementary Fig. 2; mean RT:  $t(1,38) = 15.16$ ,  $\beta = 0.39$ , 95% CI  $[.33 .44]$  and error:  $t(1,38) = 13.43$ ,  $\beta = 0.87$ , 95% CI  $[0.7 1.0]$ ). These effects persisted across sessions without substantial changes, mean RT:  $t(1,38) = -1.64$ ,  $\beta = 0.04$ , 95% CI  $[-0.09 0.06]$  for linear effect;  $t(1,38) = 1.48$ ,  $\beta = 0.03$ , 95% CI  $[-0.01 0.06]$  for quadratic effect, and error:  $t(1,38) = -1.99$ ,  $\beta = 0.87$ , 95% CI  $[0.7 1.0]$  for linear effect;  $t(1,38) = 1.40$ ,  $\beta = 0.13$ , 95% CI  $[-0.05 0.30]$  for quadratic effect. A direct comparison of complete repetition trials and all-switching trials also did not significantly change across days [ $t(1,38) < 1.50$ ].

However, although the overall benefits of S-R repetitions compared to switching did not change significantly across days [ $t(1,38) < 0.16$ ], the cost on RT of switching rules in S-R repetitions trials was significantly reduced across days, mean RT:  $t(1,38) = -4.12$ ,  $\beta = -0.03$ , 95% CI  $[-0.05 -0.01]$  for linear effect;  $t(1,38) = 2.71$ ,  $\beta = 0.03$ , 95% CI  $[0.01 0.04]$  for quadratic effect, and error:  $t(1,38) = 0.97$ ,  $\beta = 0.07$ , 95% CI  $[-0.06 0.20]$  for linear effect;  $t(1,38) = -0.64$ ,  $\beta = -0.05$ , 95% CI  $[-0.02 0.08]$  for quadratic effect. These results suggest that learning across days reduced the costs of switching action rules most substantially.

Finally, we tested how trial-to-trial changes in action-relevant but redundant features affected performance. Specifically, two different cues signaled the same task rule. Thus, while using the cue to identify the task was important, differentiating between these cues was not critical to solving the task. Nevertheless, consistent with past observations of cue encoding effects (Arrington et al. 2007), when the task cue changed from one trial to the next, but all other task features (ie the rule, stimulus, and response) remained identical, costs in RTs, errors, and variability of RTs were evident (Fig. 1C; mean RT:  $t(1,38) = 11.18$ ,  $\beta = 0.67$ , 95% CI  $[0.55 0.79]$ ; Error:  $t(1,38) = 7.15$ ,  $\beta = 0.77$ , 95% CI  $[0.56 0.99]$ ; variability of RT,  $t(1,38) = 11.18$ ,  $\beta = 0.67$ , 95% CI  $[0.56 0.79]$ ). Such cue-dependent switch costs were significantly reduced across days for the mean RTs and variability of RTs (mean RT:  $t(1,38) = -3.43$ ,  $\beta = -0.36$ , 95% CI  $[-0.57 -0.15]$  for linear effect, and  $t(1,38) = 0.71$ ,  $\beta = 0.07$ , 95% CI  $[-0.13 0.29]$  for quadratic effect; error:  $t(1,38) = -1.47$ ,  $\beta = -0.28$ , 95% CI  $[-0.65 0.09]$  for linear effect, and  $t(1,38) = 1.81$ ,  $\beta = 0.34$ , 95% CI  $[-0.03 0.70]$  for quadratic effect; Variability of RT:  $t(1,38) = -3.48$ ,  $\beta = -0.36$ , 95% CI  $[-0.57 -0.15]$  for linear effect, and  $t(1,38) = 0.71$ ,  $\beta = 0.07$ , 95% CI  $[-0.13 0.28]$  for quadratic effect). Thus, aligned with the improvement across days in general task performance and switching of action rules, action selection became more resilient to differences in task features that are redundant to the objective of the task.

### Task representations at an early stage of learning (day 1)

As session one (day 1) showed strong power law-like improvement within the session, we first focused on analysis within this session. Over the course of action selection, we characterized how task representations of task rules, stimuli, responses, S-R conjunctions, and context-specific S-R conjunctions (ie Rule S-R conjunction) change along with the improvement of task performance via single-trial RSA (Fig. 1C).

Temporal trajectories of task representations closely followed the pattern observed previously (Kikumoto and Mayr 2020; Fig. 3A). During the cue-to-stimulus interval, cued rules were encoded. Then, during action selection after the stimulus onset, context-specific conjunctive subspaces emerged that are dissociable from other lower-dimensional task representations. All of these events preceded response execution (Supplementary Fig. 3), replicating the previous studies (Kikumoto and Mayr 2020; Kikumoto, Bhandari, Shibata & Badre, in press).

Next, we directly tested how accumulated instances of specific action contexts changed the strength of encoded task representations. We found that as the function of the logarithmic increase of the number of instances or episodes of successful context-specific actions, the strength of context-specific S-R conjunctions increased at the early stage of action selection (ie 200 to 300 ms after the onset of stimulus; Fig. 3B). This effect on context-specific conjunctions remained significant after controlling for trial-to-trial RT as a nuisance predictor,  $t(1,38) = 2.38$ ,  $\beta = 0.02$ , 95% CI  $[0.01 0.05]$ . A similar effect of the number of instances was observed for encoding stimulus and response representations in the late response selection period (Fig. 3B); however, such effects were not observed for S-R conjunctions.

We further found that enhancement of context-specific conjunctions had a significant effect on how rapidly performance improvement progressed on day 1. Specifically, stronger context-specific conjunctions were associated with having fewer instances of trials within specific action contexts leading to the asymptotes in mean RT,  $t(1,38) = 2.19$ ,  $\beta = 0.01$ , 95% CI  $[0.01$

**Table 2.** MLM testing the effects of task representations (0 to 300 ms from stimulus onsets) on RTs across sessions, focusing on the early phase (first 75 blocks).

| Effects             | Rule             |         | Stimulus          |         | Response          |         | S-R Conjunction   |         | Rule-specific Conjunction |         |
|---------------------|------------------|---------|-------------------|---------|-------------------|---------|-------------------|---------|---------------------------|---------|
|                     | beta (CI)        | t-value | beta (CI)         | t-value | beta (CI)         | t-value | beta (CI)         | t-value | beta (CI)                 | t-value |
| Main effect         | .01 (-0.01 .02)  | .93     | -.02 (-0.03 -.01) | -2.21   | -.02 (-0.04 -.01) | -2.58   | -.03 (-0.05 -.01) | -2.72   | .02 (-0.04 -.01)          | -3.21   |
| Session (linear)    | -.03 (-0.06 .01) | -1.98   | -.01 (-0.04 .03)  | -.28    | -.01 (-0.04 .02)  | -0.63   | -.01 (-0.05 .01)  | -.89    | -.01 (-0.04 .02)          | -.84    |
| Session (quadratic) | .01 (-0.02 .05)  | .94     | -.03 (-0.06 -.01) | -2.23   | -.05 (-0.05 .01)  | .55     | -.02 (-0.05 .11)  | -1.22   | -.02 (-0.05 .01)          | -1.74   |

0.07] for the interaction effect, though not in the variability of RT,  $t(1,38)=0.65$ ,  $\beta = <0.01$ , 95% CI [-0.02 0.03]. No such relationship was significant for other task representations. Thus, these results indicate that encoding context-specific conjunctions is enhanced by accumulated experiences in specific action contexts on day 1, and this accumulation is directly related to classical power-law improvements in performance within the early session.

### Changes in task representations across days

Next, we considered how representations changed over the course of sessions (days 1, 2, and 3). Across days, we observed that task representations at specific stages of action selection were selectively enhanced (Fig. 3A). During the cue-stimulus interval, the strength of rule representations remained the same across days,  $t(1,38)=0.18$ ,  $\beta < 0.01$ , 95% CI [-0.01 0.01] for linear effect and  $t(1,38)=-1.54$ ,  $\beta = -0.01$ , 95% CI [-0.01 0.01] for quadratic effect. In contrast, during an early stage of response selection (0 to 300 ms after the stimulus onset), rule S-R conjunctions were significantly enhanced, along with stimulus and response representations (Table 1). Importantly, these changes over days were absent for rule representations, and the strength of S-R conjunctions significantly decreased across days within this period (Fig. 3A). However, this decrease in S-R conjunctions was driven by shorter durations of activation from the stimulus onset, and the strength remained similar at the time responses were executed (Supplementary Fig. 3). In fact, their strength increased briefly after response execution. These results indicate that each task representation is optimized at variable rates of change while participants gain expertise.

Although most task representations, in particular context-specific S-R conjunctions, were consistently negatively correlated with trial-to-trial RTs across all sessions (Supplementary Fig. 4), we did not observe significant interactions between behavioral changes across sessions and the strength of moment-by-moment RSA scores, (Rule S-R conjunction:  $t(1,38)=-0.84$ ,  $\beta = -0.01$ , 95% CI [0.04 0.02] for linear effect;  $t(1,38)=-0.84$ ,  $\beta = -0.01$ , 95% CI [0.04 0.02] for quadratic effect; S-R conjunction:  $t(1,38)=-0.89$ ,  $\beta = -0.06$ , 95% CI [-0.05 0.02] for linear effect;  $t(1,38)=-1.22$ ,  $\beta = -0.02$ , 95% CI [-0.05 0.01] for quadratic effect). Further exploratory analyses also confirmed that the effects are absent even when focused only on the early phase of each session where the influence of fatigue-like patterns in later sessions is minimal (Table 2). Thus, unlike the within-session changes in performance at was tied to the accumulation of instances, we did not find evidence that offline gains are due to factors affecting decodability of task representations.

### Changes in representational dimensionality across days

Further, we assessed how practice changes the overall representational dimensionality of the task input space (ie unique action contexts) across days and how their changes relate to the quality of individual task representations. We previously observed

that the higher-dimensional formats are associated with faster and accurate responses, which peaked around 100 ms before response execution (Kikumoto et al. 2024). Across days, a similar temporal trajectories were replicated; however, the dimensionality only marginally increased at the peak across sessions (Fig. 4A;  $t(1,38)=1.45$ ,  $\beta = 0.18$ , 95% CI [-0.06 0.41] for the linear effect;  $t(1,38)=-0.08$ ,  $\beta = 0.01$ , 95% CI [-0.25 0.28] for the quadratic effect). However, this increase significantly correlated with encoding of stronger context-specific conjunctions (Fig. 4B). No such changes across sessions were observed for other task representations, including stimulus and response conjunctions. As an exploratory analysis, we correlated the changes in the variability in RSA scores of all task representations with the changes in dimensionality estimates across individual participants. We observed no significant correlations ( $t < 0.86$ ).

Both the increase of signal-to-noise ratio in encoding of context-specific conjunctions and the addition of diverse mixing of task features (ie, heterogeneous mixing) are known to expand the representational dimensionality (Fusi et al. 2016; Kikumoto et al. 2024). During dimensionality estimation, the exclusive cutoff method introduced by Kikumoto et al. (2024) effectively prevents potential inflation of dimensionality estimates driven by changes in the signal-to-noise ratio (ie variability) of constituent task factors such as rule, stimulus, and response. Thus, our method provides a conservative, but minimally biased, chance to detect the increase in dimensionality driven by adding unique coding axes. Taken together, these results suggest that practice did not add a heterogeneous mixing of task features as additional representations; rather, it enhanced, task-tailored, context-specific conjunctions (but see [Changes in Task-redundant Dimensions](#)).

### Changes in representational dynamics across days

We previously observed that reaching stable neural dynamics in conjunctive subspaces is an important precondition for efficient context-dependent action selection (Kikumoto, Bhandari, Shibata & Badre, in press). If practice leads to better access to task memories providing direct solutions, the neural dynamics leading to orthogonal subspaces of goal states of the task (ie context-specific conjunctions) should be reshaped to be more efficient to provide a stable readout for outputs. Specifically, the stable subspaces should be established earlier during action selection, and the subsequent readout should be resilient to task features that are not informative to define required goal states.

To test this hypothesis, we compared how the temporal stability of context-specific conjunctions changed over practice, focusing on early response selection (0 to 300 ms after the stimulus onset) where earlier entries to stable subspaces should be most pronounced. We found that the temporal stability of context-specific conjunctions increased across days (Fig. 5). From day 1 to day 2, the stability generally increased during early response selection (0 to 300 ms after the stimulus onset). A similar yet more pronounced effect was observed when day 1 to day 3 were compared. These results suggest that the neural patterns during

**Table 3.** MLM testing the effects of sessions (days) on cross-cue generalization (0 to 300 ms from the stimulus onset).

| Effects             | Rule           |         | Stimulus        |         | Response        |         | S-R Conjunction |         | Rule-specific Conjunction |         |
|---------------------|----------------|---------|-----------------|---------|-----------------|---------|-----------------|---------|---------------------------|---------|
|                     | beta (CI)      | t-value | beta (CI)       | t-value | beta (CI)       | t-value | beta (CI)       | t-value | beta (CI)                 | t-value |
| Session (linear)    | .04 (-.14-.22) | .48     | .16 (.05-.28)   | 2.91    | .44 (.26-.63)   | 4.75    | -.42 (-.62-.21) | -4.1    | .23 (.05-.04)             | 2.56    |
| Session (quadratic) | .13 (-.05-.32) | .12     | -.02 (-.14-.08) | -.01    | -.16 (-.34-.03) | -1.65   | -.01 (-.01-.01) | -1.23   | .23 (-.41-.05)            | -2.46   |

action selection become more stable at an earlier timing relative to the onset of stimulus (Table 3). Notably, from day 2 to day 3, the forward generalization (ie the upper-left portion of the generalization matrix) is further enhanced, yet some effects overlapped with timing where responses were executed.

Focusing on early time windows where session-wise changes were detected, more stable conjunctive subspaces were correlated with faster trial-to-trial RTs,  $t(1,38) = -2.85$ ,  $\beta = -0.02$ , 95% CI  $[-0.28 \text{ } -0.01]$ , controlling for the effect of time-resolved strength of conjunctions,  $t(1,38) = -3.40$ ,  $\beta = -0.02$ , 95% CI  $[-0.36 \text{ } -0.01]$  or other task representations, replicating prior observations (Kikumoto, Bhandari, Shibata & Badre, in press). Further, the stable component showed a marginally significant interaction correlating with changes in RT across sessions,  $t(1,38) = -1.82$ ,  $\beta = -0.02$ , 95% CI  $[-0.38 \text{ } 0.01]$ , such that more evidence of stable dynamics predicting better improvement across days. In contrast, the dynamic component showed a marginally significant interaction in the opposite direction,  $t(1,38) = 2.05$ ,  $\beta = -0.02$ , 95% CI  $[0.09 \text{ } 0.04]$ , meaning that more evidence of dynamic representations predicted less improvement across days; however, this difference between the stable and dynamic component on RTs across sessions was not statistically significant,  $t(1,38) = 1.49$ ,  $\beta = 0.018$ , 95% CI  $[-0.01 \text{ } 0.04]$ . Thus, these results indicate that the neural dynamics that reach stable conjunctive subspaces are enhanced by extensive practice, which leads to better task performance.

### Changes over task-redundant dimensions

Finally, we tested whether optimal task representations are formed not only by separating context-specific conjunctive subspaces but also by compressing redundant dimensions within rule contexts. Specifically, we tested the degree of abstraction by characterizing alignment across task-redundant cues (Fig. 1B).

As noted above, switching task cues incurs significant costs, even when the rule context and S-R mapping stay identical (Meiran 1996; Mayr 2006; Jost et al. 2013). These costs were reduced over practice (Fig. 6B; see Practice Effects on Behavior). We further found that, across days, the context-specific conjunctions become more generalizable (ie abstract) across cues in both the dynamic and stable components of the dynamics (Fig. 6B),  $t(1,38) = 2.56$ ,  $\beta = 0.23$ , 95% CI  $[0.05 \text{ } 0.41]$  for linear effect and  $t(1,38) = -2.46$ ,  $\beta = -0.22$ , 95% CI  $[-0.41 \text{ } -0.05]$  for quadratic effect (see Table 3 for other task features), suggesting practice over days reorganized the dynamics to be similar across task cues.

Further, these changes were behaviorally relevant. In participants showing more generalized patterns of context-specific conjunctions, the costs of RTs and the variability of RTs tended to be lower (Fig. 6C; see Supplementary Fig. 4 for other task features; RT cue-switch costs:  $t(1,38) = -4.26$ ,  $\beta = -0.28$ , 95% CI  $[-0.41 \text{ } -0.15]$ ; SD cue-switch costs:  $t(1,38) = -4.62$ ,  $\beta = -0.30$ , 95% CI  $[-0.43 \text{ } -0.17]$ ). Further, such differences among individuals become progressively stronger across days (RT cue-switch costs:  $t(1,38) = -2.87$ ,  $\beta = -0.39$ , 95% CI  $[-0.66 \text{ } -0.12]$  for linear effect;  $t(1,38) = -2.07$ ,  $\beta = -0.26$ , 95% CI  $[-0.49 \text{ } -0.02]$  for quadratic effect; SD cue-switch costs:  $t(1,38) = -4.62$ ,  $\beta = -0.30$ , 95% CI  $[-0.50 \text{ } -0.01]$  for linear effect;  $t(1,38) = -1.74$ ,  $\beta = -0.20$ , 95%

CI  $[-0.43 \text{ } 0.26]$  for linear effect). Taken together, these results indicate that extensive practice reshaped the neural dynamics encoding context-specific conjunctions to be more stable and abstract, providing more efficient access to the subspace corresponding to the minimally sufficient task-tailored solutions.

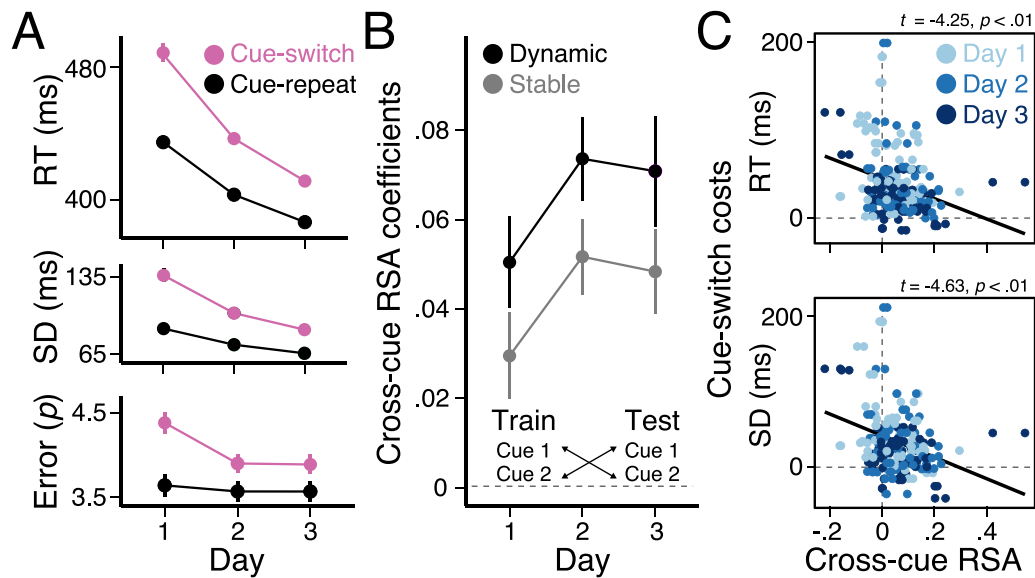
### Discussion

What changes in neural task representations lead to performance gains over the course of extensive practice? We report two results that give novel insights into this question. First, we found that accumulated experience with specific task instances over the course of a session—defined in terms of a particular co-occurrence of stimulus, response, and rule context—led to the strengthening of high-level context-specific conjunctive representations. Over the course of practice, the dynamics encoding these representations became stabilized at progressively earlier points in time with respect to the beginning of trials, and they contributed to increased dimensionality of task representations. The strengthening of these conjunctive representations, and not other levels of representation, accounted for the power law-like improvement in performance during the early phase of learning. Second, we observed that while conjunctions were strengthened, they were also shaped over days so that dynamics became generalizable (ie aligned) over redundant task cues. These changes were correlated with reductions in the costs of switching task cues. Taken together, our results indicate that practice optimizes stable encoding and retrieval of task-tailored context-specific conjunctive representations. We now consider each finding in turn.

On each trial of the task, participants encountered a specific combination of rule, stimulus, and response. As experience with these trial events accumulated over the course of a session, we observed that nonlinear conjunctive representations of the combination of these features increased in strength with each encounter (Fig. 3B). We also saw a similar increase in strength of the stimulus and response representations with experience, but critically not for S-R conjunctions. However, only the changes in context-specific conjunctions accounted for the power law-like rapid improvement in performance in the first session.

Different theories of skill learning, habits, and automatic behavior have emphasized learning at different levels of task representation. A long tradition in the study of cognitive control, for instance, has emphasized the learning of context-independent S-R mappings as the basis of automatic behavior (Miller and Cohen 2001). Information about rules and contexts maintained in working memory can bias the activation of task-relevant S-R associations over competitors, which serves as the basis for controlled behavior. Similarly, several models of motor learning assume learning of S-R mappings is at the base of practice-based change to gain efficiency (Wood and R  nger 2016; Hardwick et al. 2019; Du et al. 2022).

Similarly, chunking theories of automaticity assume that the benefit of learning occurs first in more general and then more specific combinations or chunks (A. Newell 1993). This would



**Fig. 6.** Cue-switching costs and cross-cue generalization. A) Costs of switching cues in trials with the repetition of task rules, stimulus, and response. B) The changes in average RSA scores of rule-specific conjunctions that generalize over cues across days. The temporal generalization RSA is performed in a cue-specific manner, and then, the results are averaged over 0 to 300-ms time periods. C) Correlations between behavioral costs of switching cues (ie the average and variability of RTs) and the cross-cue generalization decoding of rule-specific conjunctive subspaces.

predict that learning of context-independent S-R conjunctions (simpler chunks) would drive performance change early in the session, while learning of context-specific conjunctions (higher-order chunks) would affect improvement late. However, we did not see changes in the S-R conjunctions that related to performance gains at any stage. This was unlikely due to our task design. Rather, our task was designed so that particular S-R mappings were encountered across different contexts. Therefore, any S-R conjunctions were experienced twice as frequently as context-specific conjunctions, offering ample opportunity for experience-based improvement. Nevertheless, we did not observe a systematic increase in the strength of these S-R conjunctions over the course of practice, nor did their strength relate to behavioral change.

In this sense, the present observations align most closely with the assumptions of instance theory (Logan 1988) that assigns learning to high-level, specific task events. According to this theory, a task can be performed initially by way of a slow but flexible algorithm that can assemble any task. However, each time a task is performed, a memory of the specific combination of task features during each processing event is obligatorily encoded as an instance in memory. On future attempts to perform the task, the slow algorithm races with the time it takes to retrieve an instance. As more instances accrue, random sampling of faster retrieval is more likely to occur, winning the race (Logan 1992). However, given a distribution of retrieval times, retrieval events that are more extreme become rarer over time, resulting in a decelerating performance improvement toward an asymptote.

In line with these ideas, our results relate practice-based improvement only to facilitated access to high-dimensional, context-specific conjunctions. Indeed, we observed changes in the neural subspaces that we have previously associated with facilitated context-dependent readout (Kikumoto and Mayr 2020; Badre et al. 2021; Kikumoto et al. 2022a; Kikumoto et al. 2022b; Rangel et al. 2023). Specifically, in prior work (Kikumoto et al., in press), we used a response-deadline procedure to characterize the representational state that facilitated correctly responding in the

virtually same task used here. Using this procedure, we found that fast and accurate task performance followed from the availability of conjunctive representations that were encoded within a high-dimensional neural geometry and that had transitioned from a temporally dynamic to a stable state.

In the present study, we found these same features improved with practice. Specifically, a stronger conjunctive representation was encoded within a higher-dimensional geometry over sessions, and this representation progressively stabilized at earlier time-points relative to the response execution. Thus, representational changes that facilitate access to the conjunctive representation to guide efficient performance are directly enhanced with repeated practice.

Over the course of days, we also observed that the conjunctive representation changed to be generalizable over task features that are salient but redundant for defining response-relevant contingencies in the task. In particular, in this task, we included two cues for each rule. For example, the vertical rule was cued by both the words “vertical” and “updown.” Early in practice, relatively independent geometries encoded conjunctions for different rule cues, even though the rule context and S-R mapping (ie context-specific conjunction) were shared. Switching these redundant task cues incurred significant costs on behavior. However, over the course of practice, these distinct conjunctive subspaces came to align, forming one conjunctive rule subspace that abstracted over differences in the redundant rule cues. This abstraction predicted a gradual reduction of behavioral cost from switching the cue.

Accordingly, in the current study, we show that high-dimensional representations that guide efficient responding are not always incidental bindings of all task-relevant features at a particular moment in time, but rather, come to represent neural states that are shaped based on abstract task demands. They incorporate only the task-relevant features needed to minimally separate task contingencies to make a response, and slowly, form abstractions over unnecessary, redundant elements. However, unlike the changes in the strength of context-specific conjunctive code, these changes to its representational geometry occurred over



the course of sessions, rather than within a session. We return to this distinction below.

It is notable that the abstract alignment of context-specific conjunctions is inconsistent with the strictest version of instance theory (Logan 1988). In its original formulation, instances were akin to episodic memory-like bindings of all features of a task event (eg task cues or timing of trials). Nonetheless, there is neither a mechanism for transforming these instances based on redundant features of the input nor a clear prediction for how the system determines the complexity of instances given the tasks. Similarly, this kind of abstraction is not a feature of chunking or other theories of automaticity. However, these theories mostly focused on improvement in power law that occurred only in session 1 in our study and so, in some sense, did not need this kind of abstraction for their explanatory scope. Thus, this observation raises the question—why does this abstraction occur?

There are several reasons abstraction of this kind would be advantageous. First, abstraction is a basis for generalization that can provide robustness against interference (eg input variability) and can allow reuse of representations for new task scenarios or transfer to new tasks. Second, coding extra dimensions may be costly in several ways, and so abstraction may be encouraged by some form of regularization (Wojcik et al. 2023). Finally, abstraction allows separate instances of action contexts to be treated as one, allowing them to benefit from repetition, which leads to performance benefits. Indeed, our data provide evidence of this last advantage, showing that abstraction over the rule cue predicted reduced cue switch costs over sessions.

Collectively, then, these results provide an answer to how task representational geometry changes with practice, and specifically to the question of whether dimensionality expands or reduces over the course of practice. Paradoxically, both kinds of change occur. There is progress toward higher-dimensional geometry driven by an enhanced access to task-tailored (ie structured, not random), context-specific conjunctions. These changes reflect greater separation among conjunctive subspaces, as well as changes in features like temporal stability—initially more pronounced during early response selection and extending to later time periods over sessions. More generally, these changes reflect more robust maintenance of task-tailored, high-dimensional geometry and optimization for better performance, possibly extending beyond completing current trials (Badre et al. 2021; Rangel et al. 2023; Kikumoto et al. 2024). However, there is also a shift toward a more efficient and compressed (ie lower-dimensional) space of conjunctions within the structure and objective of task contingencies.

Thus, rather than conceptualizing automaticity as progress toward higher or lower dimensional task representations. Our results suggest that practice optimizes access to a tailored task representation that incorporates combinations of all those features, and only those features, needed to specify a correct output to solve each instance of the task at hand. Thus, in time, this accessible, high-dimensional code features a library of specific task events that tessellate the full task space (as defined by the task's objective function) but not necessarily the full input space.

A key open question concerns what mechanisms of learning underlie the changes we observe and why. The pattern of optimization we observe here is in line with what would characterize representations found in neural network models trained to solve a context-dependent response selection task (Farrell et al. 2022). If learning abstract dynamics reduces large costs in task performance (Fig. 6), then gradient-based optimization should

discover such solution manifolds (Saxe et al. 2022; Goudar et al. 2023; Driscoll et al. 2024; Sandbrink and Summerfield 2024). However, nonlocal gradient-based methods (eg backpropagation) and their approximations may not accurately represent what occurs during extensive practice, nor are they necessarily desirable (Scott and Frank 2024). Nevertheless, plausible biological mechanisms, such as bidirectional activation recirculation (Movellan 1991; Oreilly 1996) or three-factor plasticity combined with noise correlation (Frémaux et al. 2013; Nassar et al. 2021), are known to support comparable optimizations that promote robustness against interference and enhance generalization.

Regardless of the types of algorithms, however, a further open issue is that optimization requires an objective function to be defined and maintained without interference from other objectives and shared within the system (Richards and Kording 2023). While this is straightforward to implement in artificial agents within well-controlled learning contexts, it is not clear how such an objective of learning is implemented in a human, even when practicing one task (eg an instructed experimental task), alongside many other tasks and objectives in various contexts (eg doing homework). Thus, our data establish that extensive practice may be understood as a process of optimizing access to task-tailored conjunctive codes across conditions of a task, but it is not clear how practice yields this change.

While our data cannot establish the mechanisms of learning, our results do offer potential clues as a basis for future research. In particular, we observed a key distinction in the time course of practice-based change between access to the conjunction versus shaping of the conjunction. Again, changes in the strength and accessibility of the conjunctive representation only explained the within-session power law-like improvement in performance. We did not see a relationship between these changes and the offline gains in performance observed for each session. Conversely, abstraction of the conjunctive subspace over the redundant task cues input unfolded over multiple sessions that were separated by long rest periods including a night of sleep. That suggests that reshaping of the dynamic geometry of task representations occurs at multiple temporal and representational scales (Kikumoto and Mayr 2018) and might utilize different mechanisms.

While abrupt, nonlinear learning (eg after rest) is ubiquitously observed in visual perceptual learning and motor learning tasks (Walker et al. 2002; Censor et al. 2012; Tamaki et al. 2020), we did not predict this difference in time course, and so it will be important to replicate this observation. Nevertheless, this difference might point at an important difference in learning. For example, changes in separation and stability that facilitate access to conjunctive representations might be driven by performance-dependent mechanisms like Hebbian plasticity in cortical synapses or episodic encoding mechanisms through the hippocampus that occur within the session (Mill and Cole 2023). In contrast, the changes leading to the abstraction over irrelevant features might require latent consolidation mechanisms, such as those associated with the formation of abstract schemas in memory (Preston and Eichenbaum 2013; Rasch and Born 2013; Behrens et al. 2018), statistical regularizations of the learned weights (Löwe et al. 2024), or replay (Preston and Eichenbaum 2013) and synaptic downscaling (Norimoto et al. 2018) that are often regulated over a night of sleep. Future work manipulating the conditions of practice or overnight consolidation might provide insights into these mechanisms of change.

There are limitations in the present study that put our results in context. A first consideration is the generalizability of the present conclusions. We tested only one task, and, as such,

we leave open the possibility that other tasks would lead to different outcomes. It will be important to replicate the present results in other tasks and paradigms, particularly in settings that are more complex, temporally extended, demanding of working memory, and/or ecologically valid. Nevertheless, the present task is designed to test how different responses are selected to the same stimulus depending on a contextual rule. That context-sensitivity sets this task apart from the S-R learning tasks commonly used in the motor learning literature (Hardwick et al. 2019). This basic context-dependent rule structure is representative of most, if not all, tasks requiring cognitive control at their most reduced level.

Further, while the task lends itself to focusing on context-dependent conjunctions, we also included the opportunity to learn more low-dimensional intermediate representations, like the common S-R mappings, that were shared over more trials and so could have facilitated behavior under some models of practice-related improvement. But, again, although S-R conjunctions were correlated with trial-to-trial performance, they were not consistently associated with practice-related change. Thus, we believe the current observations will generalize to many other tasks and reflect general properties of practice-related change in goal-directed behavior requiring context-sensitivity.

While we view our observations to be generalizable to other cases of practice of a stable, context-dependent rule structure, we have not tested all aspects of practice and skill learning, and so, we have not identified the only drivers of practice-related change in performance. For example, changes in task rules and strategies (eg discovering that going over the bar backward improves the high jump) will result in abrupt shifts in task performance (Taylor and Ivry 2012; Schuck et al. 2015). Such changes can be discovered through experience, and our results do not address such effects. Rather, we view these observations as relating to refinement and optimization given a particular stable strategy, objective, and structure of the task, thereby expressing how target representational states change to facilitate their access and match to task contingencies over the course of practice.

In summary, we observe that experience with a task enhances access to context-integrated conjunctive representations that correspond to specific combinations of rule, stimulus, and response during each task event within a task space, as defined by a task's objective function. This facilitated access accounted for the power law-like improvements observed within a session of practice. Over the course of multiple sessions, this target conjunctive representation was shaped to be encoded in temporally stable dynamics and abstract neural subspaces. These results provide a new window on the targets of practice-related change that leads to performance improvement on a task.

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## Author contributions

Project conception: A.K. and D.B.; Methodology: A.K.; Data Collection: A.K., T.N., and K.S.; Analysis and Interpretation: A.K. and D.B.; Manuscript writing: A.K., K.S., and D.B.; Funding acquisition:

K.S. and D.B.; Supervision: K.S. and D.B. Atsushi Kikumoto (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Validation, Visualization, Writing—original draft, Writing—review & editing), Kazuhisa Shibata (Funding acquisition, Writing—review & editing), Takahiro Nishio (Data curation), David Badre (Conceptualization, Funding acquisition, Writing—original draft, Writing—review & editing).

## Supplementary material

Supplementary material is available at *Cerebral Cortex* online.

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## References

- Anderson JR. 1982. Acquisition of cognitive skill. *Psychol Rev.* 89: 369–406. <https://doi.org/10.1037/0033-295X.89.4.369>.
- Arrington CM, Logan GD, Schneider DW. 2007. Separating cue encoding from target processing in the explicit task-cuing procedure: are there “true” task switch effects. *J Exp Psychol Learn Mem Cogn.* 33:484–502. <https://doi.org/10.1037/0278-7393.33.3.484>.
- Badre D, Bhandari A, Keglövits H, Kikumoto A. 2021. The dimensionality of neural representations for control. *Curr Opin Behav Sci.* 38: 20–28. <https://doi.org/10.1016/j.cobeha.2020.07.002>.
- Behrens TEJ et al. 2018. What is a cognitive map? Organizing knowledge for flexible behavior. *Neuron.* 100:490–509. <https://doi.org/10.1016/j.neuron.2018.10.002>.
- Bernardi S et al. 2020. The Geometry of Abstraction in the Hippocampus and Prefrontal Cortex. *Cell.* 183:954–967.e21. <https://doi.org/10.1016/j.cell.2020.09.031>.
- Bhandari A, Keglövits H, Badre D. 2024. Task structure tailors the geometry of neural representations in human lateral prefrontal cortex. *BioRxiv*. The Preprint Server for Biology. <https://doi.org/10.1101/2024.03.06.583429>.
- Censor N, Sagi D, Cohen LG. 2012. Common mechanisms of human perceptual and motor learning. *Nat Rev Neurosci.* 13:658–664. <https://doi.org/10.1038/nrn3315>.
- Cohen MX. 2014. *Analyzing neural time series data: Theory and practice.* MIT Press.
- Cohen JD, Dunbar K, McClelland JL. 1990. On the control of automatic processes: a parallel distributed processing account of the Stroop effect. *Psychol Rev.* 97:332–361. <https://doi.org/10.1037/0033-295X.97.3.332>.
- Cole MW, Patrick LM, Meiran N, Braver TS. 2018. A role for proactive control in rapid instructed task learning. *Acta Psychol.* 184:20–30. <https://doi.org/10.1016/j.actpsy.2017.06.004>.
- Courellis HS et al. 2023. Abstract representations emerge in human hippocampal neurons during inference behavior. *BioRxiv*. The Preprint Server for Biology. <https://doi.org/10.1101/2023.11.10.566490>.

- Crossman ERFW. 1959. A theory of the acquisition of speed-skill. *Ergonomics*. 2:153–166. <https://doi.org/10.1080/00140135908930419>.
- Dehaene S, King J-R. 2016. Decoding the dynamics of conscious perception: The temporal generalization method. In *Research and Perspectives in Neurosciences* (pp. 85–97). Springer International Publishing, [https://doi.org/10.1007/978-3-319-28802-4\\_7](https://doi.org/10.1007/978-3-319-28802-4_7), Cham.
- Delorme A, Makeig S. 2004. EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *J Neurosci Methods*. 134:9–21. <https://doi.org/10.1016/j.jneumeth.2003.10.009>.
- Driscoll LN, Shenoy K, Sussillo D. 2024. Flexible multitask computation in recurrent networks utilizes shared dynamical motifs. *Nat Neurosci*. 27:1349–1363. <https://doi.org/10.1038/s41593-024-01668-6>.
- Du Y, Krakauer JW, Haith AM. 2022. The relationship between habits and motor skills in humans. *Trends Cogn Sci*. 26:371–387. <https://doi.org/10.1016/j.tics.2022.02.002>.
- Duncker L, Sahani M. 2021. Dynamics on the manifold: Identifying computational dynamical activity from neural population recordings. *Curr Opin Neurobiol*. 70:163–170. <https://doi.org/10.1016/j.conb.2021.10.014>.
- Farrell M, Recanatesi S, Moore T, Lajoie G, Shea-Brown E. 2022. Gradient-based learning drives robust representations in recurrent neural networks by balancing compression and expansion. *Nat Mach Intell*. 4:564–573. <https://doi.org/10.1038/s42256-022-00498-0>.
- Farrell M, Recanatesi S, Shea-Brown E. 2023. From lazy to rich to exclusive task representations in neural networks and neural codes. *Curr Opin Neurobiol*. 83:102780. <https://doi.org/10.1016/j.conb.2023.102780>.
- Fine JM, Johnston WJ, Yoo SBM, Ebitz RB, Hayden BY. 2022. Subspace orthogonalization as a mechanism for binding values to space. *ArXiv [q-Bio.NC]*. arXiv. <https://doi.org/10.48550/ARXIV.2205.06769>.
- Flesch T, Juechems K, Dumbalska T, Saxe A, Summerfield C. 2022. Orthogonal representations for robust context-dependent task performance in brains and neural networks. *Neuron*. 110:4212–4219. <https://doi.org/10.1016/j.neuron.2022.12.004>.
- Frémaux N, Sprekeler H, Gerstner W. 2013. Reinforcement learning using a continuous time actor-critic framework with spiking neurons. *PLoS Comput Biol*. 9:e1003024. <https://doi.org/10.1371/journal.pcbi.1003024>.
- Frings C et al. 2020. Binding and retrieval in action control (BRAC). *Trends Cogn Sci*. 24:375–387. <https://doi.org/10.1016/j.tics.2020.02.004>.
- Fusi S, Miller EK, Rigotti M. 2016. Why neurons mix: high dimensionality for higher cognition. *Curr Opin Neurobiol*. 37:66–74. <https://doi.org/10.1016/j.conb.2016.01.010>.
- Goudar V, Peysakhovich B, Freedman DJ, Buffalo EA, Wang X-J. 2023. Schema formation in a neural population subspace underlies learning-to-learn in flexible sensorimotor problem-solving. *Nat Neurosci*. 26:879–890. <https://doi.org/10.1038/s41593-023-01293-9>.
- Hardwick RM, Forrence AD, Krakauer JW, Haith AM. 2019. Time-dependent competition between goal-directed and habitual response preparation. *Nat Hum Behav*. 3:1252–1262. <https://doi.org/10.1038/s41562-019-0725-0>.
- Hastie T, Buja A, Tibshirani R. 1995. Penalized Discriminant Analysis. *Ann Stat*. 23:73–102. <https://doi.org/10.1214/aos/1176324456>.
- Hastie T et al. 2024. *mda: Mixture and Flexible Discriminant Analysis (Version 0.5-5) [R package]*. CRAN. <https://CRAN.R-project.org/package=mda>.
- Hazy TE, Frank MJ, O'Reilly RC. 2007. Towards an executive without a homunculus: Computational models of the prefrontal cortex/basal ganglia system. *Philos Trans R Soc Lond Ser B Biol Sci*. 362:1601–1613. <https://doi.org/10.1098/rstb.2007.2055>.
- Hommel B. 2004. Event files: feature binding in and across perception and action. *Trends Cogn Sci*. 8:494–500. <https://doi.org/10.1016/j.tics.2004.08.007>.
- Hummos A, Wang BA, Drammis S, Halassa MM, Pleger B. 2022. Thalamic regulation of frontal interactions in human cognitive flexibility. *PLoS Comput Biol*. 18:e1010500. <https://doi.org/10.1371/journal.pcbi.1010500>.
- Ito T, Yang GR, Laurent P, Schultz DH, Cole MW. 2022. Constructing neural network models from brain data reveals representational transformations linked to adaptive behavior. *Nat Commun*. 13:673. <https://doi.org/10.1038/s41467-022-28323-7>.
- Jazayeri M, Ostojic S. 2021. Interpreting neural computations by examining intrinsic and embedding dimensionality of neural activity. *Curr Opin Neurobiol*. 70:113–120. <https://doi.org/10.1016/j.conb.2021.08.002>.
- Johnston WJ, Fusi S. 2023. Abstract representations emerge naturally in neural networks trained to perform multiple tasks. *Nat Commun*. 14:1–18. <https://doi.org/10.1038/s41467-023-36583-0>.
- Jost K, De Baene W, Koch I, Brass M. 2013. A review of the role of cue processing in task switching. *Z Psychol*. 221:5–14. <https://doi.org/10.1027/2151-2604/a000125>.
- Karni A, Tanne D, Rubenstein BS, Askenasy JJ, Sagi D. 1994. Dependence on REM sleep of overnight improvement of a perceptual skill. *Science*. 265:679–682.
- Kikumoto A, Mayr U. 2018. Decoding hierarchical control of sequential behavior in oscillatory EEG activity. *elife*. 7. <https://doi.org/10.7554/eLife.38550>.
- Kikumoto A, Mayr U. 2020. Conjunctive representations that integrate stimuli, responses, and rules are critical for action selection. *Proc Natl Acad Sci USA*. 117:10603–10608. <https://doi.org/10.1073/pnas.1922166117>.
- Kikumoto A, Mayr U, Badre D. 2022a. The role of conjunctive representations in prioritizing and selecting planned actions. *elife*. 11. <https://doi.org/10.7554/eLife.80153>.
- Kikumoto A, Sameshima T, Mayr U. 2022b. The Role of Conjunctive Representations in Stopping Actions. *Psychol Sci*. 33:325–338. <https://doi.org/10.1177/09567976211034505>.
- Kikumoto A, Bhandari A, Shibata K, Badre D. 2024. A transient high-dimensional geometry affords stable conjunctive subspaces for efficient action selection. *Nat Commun*. 15:1–16. <https://doi.org/10.1038/s41467-024-52777-6>.
- King J-R, Dehaene S. 2014. Characterizing the dynamics of mental representations: The temporal generalization method. *Trends Cogn Sci*. 18:203–210. <https://doi.org/10.1016/j.tics.2014.01.002>.
- Kriegeskorte N, Kievit RA. 2013. Representational geometry: integrating cognition, computation, and the brain. *Trends Cogn Sci*. 17:401–412. <https://doi.org/10.1016/j.tics.2013.06.007>.
- Kuhn M. 2013. Predictive Modeling with R and the caret Package. *User Model User-Adap Inter*. [http://www.r-project.org/nosvn/conferences/user-2013/Tutorials/kuhn/user\\_caret\\_2up.pdf](http://www.r-project.org/nosvn/conferences/user-2013/Tutorials/kuhn/user_caret_2up.pdf).
- Li Q, Sorscher B, Sompolinsky H. 2024. Representations and generalization in artificial and brain neural networks. *Proc Natl Acad Sci USA*. 121:e2311805121. <https://doi.org/10.1073/pnas.2311805121>.
- Logan GD. 1988. Toward an instance theory of automatization. *Psychol Rev*. 95:492–527. <https://doi.org/10.1037/0033-295X.95.4.492>.
- Logan GD. 1992. Shapes of reaction-time distributions and shapes of learning curves: a test of the instance theory of automaticity. *J Exp Psychol Learn Mem Cogn*. 18:883–914. <https://doi.org/10.1037/0278-7393.18.5.883>.



- Löwe AT, Petzka M, Tzegka M, Schuck NW. 2024. N2 Sleep Inspires InsightIn bioRxiv. <https://doi.org/10.1101/2024.06.24.600359>.
- Mackay DG. 1982. The problem of flexibility, fluency, and speed-accuracy tradeoff in skilled behavior. *Psychol Rev.* 89:483–506. <https://doi.org/10.1037/0033-295X.89.5.483>.
- Maris E, Oostenveld R. 2007. Nonparametric statistical testing of EEG- and MEG-data. *J Neurosci Methods.* 164:177–190. <https://doi.org/10.1016/j.jneumeth.2007.03.024>.
- Mayr, U. 2006. What matters in the cued task-switching paradigm: tasks or cues? *Psychon Bull Rev.* 13:794–799. <https://doi.org/10.3758/BF03193999>.
- Mayr U, Bryck RL. 2005. Sticky rules: integration between abstract rules and specific actions. *J Exp Psychol Learn Mem Cogn.* 31: 337–350. <https://doi.org/10.1037/0278-7393.31.2.337>.
- Meiran N. 1996. Reconfiguration of processing mode prior to task performance. *J Exp Psychol Learn Mem Cogn.* 22:1423–1442. <https://doi.org/10.1037/0278-7393.22.6.1423>.
- Meyer DE, Kieras DE. 1997. A computational theory of executive cognitive processes and multiple-task performance: Part I. Basic mechanisms. *Psychol Rev.* 104:3–65. <https://doi.org/10.1037/0033-295X.104.1.3>.
- Mill RD, Cole MW. 2023. Neural representation dynamics reveal computational principles of cognitive task learningIn bioRxiv (p. 2023 June 27.546751). <https://doi.org/10.1101/2023.06.27.546751>.
- Miller EK, Cohen JD. 2001. An integrative theory of prefrontal cortex function. *Annu Rev Neurosci.* 24:167–202. <https://doi.org/10.1146/annurev.neuro.24.1.167>.
- Moors A, De Houwer J. 2006. Automaticity: a theoretical and conceptual analysis. *Psychol Bull.* 132:297–326. <https://doi.org/10.1037/0033-2909.132.2.297>.
- Mosteller F, Tukey JW. 1968. Data analysis, including statistics. In: Lindzey G, Aronson E, editors. *The Handbook of Social Psychology* (Vol. 2, 2nd ed.). Addison-Wesley. p. 80–203.
- Movellan JR (1991). Contrastive Hebbian Learning in the Continuous Hopfield Model. In: Touretzky DS, Elman JL, Sejnowski TJ, Hinton GE, editors. *Connectionist Models*. Elsevier. p. 10–17. <https://doi.org/10.1016/B978-1-4832-1448-1.50007-XACM>.
- Muhle-Karbe PS et al. 2023. Goal-seeking compresses neural codes for space in the human hippocampus and orbitofrontal cortex. *Neuron.* 111:3885–3899.e6. <https://doi.org/10.1016/j.neuron.2023.08.021>.
- Musslick S, Saxe A, Novick A, Reichman D, Cohen JD. 2020. On the rational boundedness of cognitive control: Shared versus separated representations. *PsyArXiv*.
- Nassar MR, Scott D, Bhandari A. 2021. Noise correlations for faster and more robust learning. *J Neurosci.* 41:6740–6752. <https://doi.org/10.1523/JNEUROSCI.3045-20.2021>.
- Neves DM, Anderson JR. 1981. Knowledge compilation: mechanisms for the automatization of cognitive skills. In: Anderson JR, editor. *Cognitive Skills and their Acquisition*. Hillsdale, NJ: Lawrence Erlbaum Associates. p. 57–84.
- Newell A. 1990. Unified theories of cognition. *The William James Lectures.* 1987:549. <https://psycnet.apa.org/fulltext/1990-98981-000.pdf>.
- Newell A. (1993). *Mechanisms of skill acquisition and the law of practice*. <https://doi.org/10.4324/9780203728178-1/mechanisms-skill-acquisition-law-practice-allen-newell-paul-rosenbloom>
- Newell A, Rosenbloom PS. 1981. Mechanisms of skill acquisition. In: Anderson JR, editor. *Cognitive skills and their acquisition*. Hillsdale, NJ: Lawrence Erlbaum Associates. p. 1–55.
- Norimoto H et al. 2018. Hippocampal ripples down-regulate synapses. *Science (New York, NY).* 359:1524–1527. <https://doi.org/10.1126/science.aao0702>.
- Ohlsson S. 1992. The learning curve for writing books: Evidence from Professor Asimov. *Psychol Sci.* 3:380–384. <https://doi.org/10.1111/j.1467-9280.1992.tb00051.x>.
- Oreilly RC. 1996. Biologically plausible error-driven learning using local activation differences: The generalized recirculation algorithm. *Neural Comput.* 8:895–938. <https://doi.org/10.1162/neco.1996.8.5.895>.
- Pang JC et al. 2023. Geometric constraints on human brain function. *Nature.* 618:566–574. <https://doi.org/10.1038/s41586-023-06098-1>.
- Preston AR, Eichenbaum H. 2013. Interplay of hippocampus and prefrontal cortex in memory. *Curr Biol.* 23:R764–R773. <https://doi.org/10.1016/j.cub.2013.05.041>.
- Rangel BO, Hazeltine E, Wessel JR. 2023. Lingering Neural Representations of Past Task Features Adversely Affect Future Behavior. *J Neurosci.* 43:282–292. <https://doi.org/10.1523/JNEUROSCI.0464-22.2022>.
- Rasch B, Born J. 2013. About sleep's role in memory. *Physiol Rev.* 93: 681–766. <https://doi.org/10.1152/physrev.00032.2012>.
- Richards BA, Kording KP. 2023. The study of plasticity has always been about gradients. *J Physiol.* 601:3141–3149. <https://doi.org/10.1113/JP282747>.
- Rigotti M, Ben Dayan Rubin D, Wang X-J, Fusi S. 2010. Internal representation of task rules by recurrent dynamics: the importance of the diversity of neural responses. *Front Comput Neurosci.* 4:24. <https://doi.org/10.3389/fncom.2010.00024>.
- Rigotti M et al. 2013. The importance of mixed selectivity in complex cognitive tasks. *Nature.* 497:1–6.
- Ritz H, Shenhav A. 2024. Orthogonal neural encoding of targets and distractors supports multivariate cognitive control. *Nat Hum Behav.* 8:945–961. <https://doi.org/10.1038/s41562-024-01826-7>.
- Sandbrink K, Summerfield C. 2024. Modelling cognitive flexibility with deep neural networks. *Curr Opin Behav Sci.* 57:101361. <https://doi.org/10.1016/j.cobeha.2024.101361>.
- Saxe AM, Sodhani S, Lewallen S (2022). The Neural Race Reduction: Dynamics of Abstraction in Gated Networks. In *arXiv [cs.LG]*. arXiv. <http://arxiv.org/abs/2207.10430>.
- Schneider W. 1985. Training high-performance skills: Fallacies and guidelines. *Hum Factors.* 27:285–300.
- Schuck NW et al. 2015. Medial prefrontal cortex predicts internally driven strategy shifts. *Neuron.* 86:331–340. <https://doi.org/10.1016/j.neuron.2015.03.015>.
- Schumacher EH, Hazeltine E. 2016. Hierarchical task representation. *Curr Dir Psychol Sci.* 25:449–454. <https://doi.org/10.1177/0963721416665085>.
- Schumacher EH et al. 2001. Virtually perfect time sharing in dual-task performance: Uncorking the central cognitive bottle-neck. *Psychol Sci.* 12:101–108. <https://doi.org/10.1111/1467-9280.00318>.
- Scott DN, Frank MJ. 2024. Beyond gradients: factorized, geometric control of interference and generalization. *Life.* 13:RP103701. <https://doi.org/10.1101/2021.11.19.466943>.
- Shiffrin RM, Schneider W. 1977. Controlled and automatic human information processing: II. Perceptual learning, automatic attending and a general theory. *Psychol Rev.* 84:127–190. <https://doi.org/10.1037/0033-295X.84.2.127>.
- Stroop JR. 1935. Studies of interference in serial verbal reactions. *J Exp Psychol.* 18:643–662. <https://doi.org/10.1037/h0054651>.
- Tafazoli S et al. 2024. Building compositional tasks with shared neural subspacesIn bioRxiv (p. 2024 January 31.578263). <https://doi.org/10.1101/2024.01.31.578263>.



- Tamaki M et al. 2020. Complementary contributions of non-REM and REM sleep to visual learning. *Nat Neurosci.* 23:1150–1156. <https://doi.org/10.1038/s41593-020-0666-y>.
- Tang E et al. 2019. Effective learning is accompanied by high-dimensional and efficient representations of neural activity. *Nat Neurosci.* 22:1000–1009. <https://doi.org/10.1038/s41593-019-0400-9>.
- Tang C, Herikstad R, Parthasarathy A, Libedinsky C, Yen S-C. 2020. Minimally dependent activity subspaces for working memory and motor preparation in the lateral prefrontal cortex. *elife.* 9. <https://doi.org/10.7554/eLife.58154>.
- Taylor JA, Ivry RB. 2012. The role of strategies in motor learning. *Ann N Y Acad Sci.* 1251:1–12. <https://doi.org/10.1111/j.1749-6632.2011.06430.x>.
- Teigen KH. 2002. One hundred years of laws in psychology. *Am J Psychol.* 115:103–118. <https://doi.org/10.2307/1423676>.
- Vaidya AR, Badre D. 2022. Abstract task representations for inference and control. *Trends Cogn Sci.* 26:484–498. <https://doi.org/10.1016/j.tics.2022.03.009>.
- Verguts T, Notebaert W. 2009. Adaptation by binding: a learning account of cognitive control. *Trends Cogn Sci.* 13:252–257. <https://doi.org/10.1016/j.tics.2009.02.007>.
- Walker MP, Brakefield T, Morgan A, Hobson JA, Stickgold R. 2002. Practice with sleep makes perfect. *Neuron.* 35:205–211. [https://doi.org/10.1016/S0896-6273\(02\)00746-8](https://doi.org/10.1016/S0896-6273(02)00746-8).
- Weber J et al. 2023. Subspace partitioning in the human prefrontal cortex resolves cognitive interference. *Proc Natl Acad Sci USA.* 120:e2220523120. <https://doi.org/10.1073/pnas.2220523120>.
- Wojcik MJ et al. 2023. Learning shapes neural geometry in the prefrontal cortex. In *bioRxiv* (p. 2023 April 24.538054). <https://doi.org/10.1101/2023.04.24.538054>.
- Wood W, R  nger D. 2016. Psychology of Habit. *Annu Rev Psychol.* 67:289–314. <https://doi.org/10.1146/annurev-psych-122414-033417>.
- Zheng Y, Lu R, Woolgar A (2024). Radical flexibility of neural representation in frontoparietal cortex and the challenge of linking it to behaviour. *CRNEUR.* 5:100043. <https://doi.org/10.1016/j.crneur.2024.100043>.